Stakeholder Question Catalogue 03

PhD Students

80 doctoral-research questions mapped to physically plausible sensing, authoritative institutional sources, explicit operation types, evidence-quality gates, uncertainty, permissions and safe answer boundaries.

STAKEHOLDER	PhD students as doctoral researchers using Abacws for long-horizon research, sustained writing, supervision and reviews, experiments, specialist computing, teaching assistance, events, presentations, irregular schedules and responsible data use.
QUESTION RANGE	Current status, historical comparison, prediction, diagnosis, recommendation, itinerary optimisation, experiment planning, evidence quality, accessibility, lone working, sustainability, privacy and safety-bounded interaction.
EVIDENCE MODEL	Proposed, correctly located sensing is combined with authorised spatial, booking, access, doctoral, BMS, energy, network, AV, equipment, maintenance, accessibility, weather, research-governance and safety data. Every record states minimum evidence, source timing, quality limits, operation type and safe failure.

Core principle: Support the doctoral decision without inspecting research content, inferring productivity, identifying occupants, substituting nearby data or sounding more certain than the evidence.

1. Catalogue scope and design principles

The catalogue focuses on decisions that PhD students may reasonably want a conversational university building to support during long-horizon research, supervision, experiments, writing, events, irregular schedules and responsible evidence use.

Proposed capability, not current fact

Every sensor, integration and service is a design requirement unless deployment, room coverage, permissions and operational status are verified. The catalogue does not claim that all capabilities currently exist.

Current building information

The May 2021 floor plan is a historical spatial reference. Current room use, ownership, student access, bookability, specialist facilities, equipment, lifts, services and HVAC relationships require authoritative verification.

Doctoral decision context

Questions reflect doctoral decisions: sustained writing and long-horizon planning; supervision and progress review; experiments and specialist computing; teaching assistance and research events; conference, publication and viva preparation; commuting, hybrid and after-hours work.

Question-to-evidence design

Each answer resolves the decision, place, interval and hard constraints; identifies minimum evidence and permissions; checks source authority, room/served-zone coverage, sampling, clock alignment, commissioning/calibration, uptime, gaps, drift and source agreement; performs the correct operation; and states uncertainty and answerability limits.

Research-content boundary

The building must not inspect drafts, code, datasets, experiments, presentations or conversations, infer productivity or monitor an individual. Only the minimum user-supplied context needed for the request is processed.

Authoritative source precedence

Bookings, access, room purpose, equipment, network, AV, closures, lone-working rules, alarms, research governance and emergency information come from authorised systems, not environmental inference.

Privacy, accessibility and safety

Use privacy-preserving aggregate occupancy and numerical acoustic statistics, never speech, identity or individual trajectories. Treat accessibility, research ethics, permissions, risk assessment, lone-working and safety constraints as hard filters.

Safe uncertainty and refusal

Forecasts are probabilistic, occupancy is an estimate, calculations expose units and coverage, diagnoses test competing explanations, and recommendations preserve hard constraints and backups. Missing, stale, below-threshold or conflicting critical evidence triggers clarification, confirmed facts only or not assessable.

2. How to read the catalogue

Readiness describes the verified sensing, institutional integrations, validation and governance needed for a responsible answer; it does not rank the academic value of the question.

R1 - practical with core feeds

Plausible once verified spatial, access, service or sensor metadata and the required core authoritative feeds are available and quality-controlled.

R2 - integration-dependent

Requires additional booking, access, doctoral, BMS, network, AV, equipment, lift, maintenance, weather or forecasting integration.

R3 - research-grade or restricted

Requires dense or specialist sensing, validated diagnosis or prediction, restricted operational or research data, or stronger privacy, ethics, security and safety governance.

Answer depth and operation types

L1 Authoritative lookup or current observation from a verified source.

L2 Calculation, aggregation or bounded historical comparison.

L3 Estimate or forecast combined with a recommendation and uncertainty.

L4 Diagnosis, optimisation, intervention evaluation or evidence audit.

Question sections

GLOBAL ANSWER RULES

Use only verified PhD-student-accessible candidates; preserve source owner/version, observed-at and retrieval times, permissions, room/served-zone mapping, sampling, calibration, uptime, gaps and drift; use anonymous occupancy with error bounds and numerical sound statistics only; distinguish observation, lookup, calculation, estimate, forecast, diagnosis and recommendation; defer access, bookings, equipment, alarms, lone-working and emergencies to authoritative systems; never inspect research content or infer productivity; and return confirmed facts only or not assessable when critical evidence fails.

3. Worked example: mixed doctoral research day

A responsible answer must combine the student-supplied schedule and hard constraints with verified access, room purpose, equipment status, room-level evidence, governance and uncertainty; it must not treat an empty or quiet space as automatically available, private or safe.

PhD student: "I will be in Abacws from 10 a.m. to 8 p.m. I need a four-hour writing block, an online supervision meeting, a short authorised experiment and a quiet evening workspace. Can you plan the campus day and tell me where evidence is too weak to recommend?"

1. Resolve the research day

Resolve times, durations, fixed locations, writing needs, call privacy, experiment authorisation, equipment, step-free access, breaks and the latest permitted evening departure.

2. Verify authorised candidates

Use current room purpose, PhD-student access, bookings, opening hours, equipment and closures; do not infer specialist capability, privacy or safety from occupancy or the 2021 plan.

3. Verify evidence and governance

Require valid room/served-zone mapping, stated sample intervals, synchronised clocks, calibration/commissioning, uptime/gap/drift flags, aggregate network/AV status, equipment service data and applicable lone-working or research-governance rules.

4. Forecast each interval

Use comparable quality-controlled periods, scheduled activity, current trends and weather; state horizon, prediction interval, omitted criteria and recheck trigger; label official lookups separately from sensor estimates and forecasts.

5. Optimise with contingencies

Apply access, experiment, privacy, accessibility and safety constraints first; minimise travel; rank primary and backup spaces; avoid backups dependent on the same unverified service.

6. Communicate answerability

State the itinerary, evidence period, confidence, omitted criteria, closing and check-in requirements, recheck points and switch triggers; return not assessable where essential evidence is absent.

ILLUSTRATIVE ANSWER STRUCTURE

"For your writing block, [verified space] is the strongest option because it meets confirmed access and service requirements and is forecast to have lower crowding and sound. Use [verified room] for the supervision call; quietness alone is not evidence of privacy. The experiment remains conditional on [authoritative approval/status]. For evening work, follow [official lone-working rule] and recheck at [time]. Confidence is [low, moderate or high] because [coverage and schedule uncertainty]. [Criterion] was omitted because [missing, stale or restricted source]."

1. Long-horizon planning, irregular schedules and focused work

Questions PHD-001-PHD-002

PHD-001

CORE | L3 - recurring-window forecast | R2 - integration-dependent

PhD student question: Over the next four weeks, I need several four-hour thesis-writing blocks. When and where am I most likely to find a quiet, comfortable workspace that I am permitted to use?

Sensors and telemetry

Room-level temperature, RH, direct NDIR CO2, task-area illuminance and locally calculated LAeq/L10/L90; anonymous room occupancy and corridor footfall. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current student-accessible space inventory, room purpose and permitted activity, opening hours, bookings, timetable/events, route graph and only the dates or constraints the student supplies. Current use and access must be verified; the May 2021 floor plan is historical spatial context, not an availability source. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified future status plus comparable quality-controlled history. Method: resolve the four-week horizon and four-hour duration; filter to verified permitted spaces; model occupancy, sound and environmental stability from comparable weeks plus future bookings/events; rank a primary and backup window with prediction intervals, evidence freshness and a recheck time. Report horizon, coverage, prediction interval and recheck trigger.

Answer boundary

This is a probability-based recommendation, not a guaranteed seat, socket or uninterrupted period. Do not infer the student's productivity or monitor their attendance. If future schedules, room-level coverage or comparable history are inadequate, provide only a broad planning range or return not assessable. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Helps the doctoral researcher protect scarce deep-work time across an irregular month while understanding that future space availability is probabilistic.

PHD-002

CORE | L2 - current recommendation | R2 - integration-dependent

PhD student question: My supervision meeting has just finished. Where can I work quietly for the next three hours without travelling far from this floor?

Sensors and telemetry

Current anonymous occupancy, room-level sound statistics, temperature, CO2 and illuminance in candidate student spaces; corridor footfall for route context; aggregate network and electrical-service status. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

The meeting location supplied by the student, current student access, room purpose, bookings, opening hours, route graph, power provision and network incident feed. The system must not read the supervision content or infer the meeting outcome. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: authoritative lookup plus calculation or bounded comparison. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: resolve the three-hour period, filter spaces by verified access and availability, calculate route distance, compare current conditions and recent trend, and rank a nearby primary and fallback space. Show primary option, backup, trade-offs and evidence time.

Answer boundary

Current occupancy does not prove a desk or socket is free, and a low sound level does not guarantee future quiet. Use only the location and timing voluntarily supplied for this request. If room status or sensing is stale, label it and avoid a precise recommendation. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Reduces transition time after supervision and helps preserve a useful block of focused work.

1. Long-horizon planning, irregular schedules and focused work

Questions PHD-003-PHD-004

PHD-003

ADVANCED | L4 - matched historical comparison | R3 - research-grade or restricted

PhD student question: Over this term, which monitored PhD workspaces have been the most consistently quiet, comfortable and well ventilated after accounting for occupancy and weather?

Sensors and telemetry

Correctly mapped room-level temperature, RH, direct NDIR CO2, task-area illuminance and locally derived LAeq/L10/L90; anonymous occupancy; outdoor temperature and solar context; BMS airflow only for the verified serving zone. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Verified current room classifications, student access, capacities, HVAC-zone relationships and timetable or event context. Do not assume that labels in the May 2021 plan remain current or that every research space is student-accessible. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: overlapping, equivalently placed streams above a declared coverage threshold. Method: define comparable spaces and periods; adjust for occupancy, time of day, weather and room type; calculate median conditions, variability, exceedance duration and data coverage; report effect sizes and uncertainty rather than a simple league table. Report effect size, coverage, uncertainty and sensitivity.

Answer boundary

Comparison is valid only where candidate spaces have equivalent sensing and enough overlapping data. Do not infer individual productivity, attendance or room entitlement. If room type, occupancy adjustment or measurement history is not comparable, report the affected result as not assessable rather than rank it. If a critical source is missing, stale, conflicting or unauthorised, restrict disclosure, provide only verified facts, or return not assessable.

Why this is useful: Supports an evidence-based choice of long-stay workspace and provides a defensible basis for raising a persistent environmental concern.

PHD-004

CORE | L3 - deadline-demand forecast | R2 - integration-dependent

PhD student question: My chapter deadline is next Friday. What arrival time gives me the best chance of finding a suitable long-stay workspace?

Sensors and telemetry

Historical anonymous occupancy, entrance and doorway counts, corridor footfall, room sound statistics and environmental measurements, with explicit sensor coverage and uptime records. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Opening hours, student-accessible room inventory, bookings, timetable, doctoral and university assessment or event calendar where authorised, planned closures and the deadline period stated by the student. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified future status plus comparable quality-controlled history. Method: identify comparable deadline and end-of-term periods, model demand by arrival time and space type, exclude closed or restricted locations, and return probability bands for finding a broadly suitable workspace plus a later fallback window. Report horizon, coverage, prediction interval and recheck trigger.

Answer boundary

Anonymous occupancy does not confirm a free desk or suitable workstation. Do not infer who has a deadline or use individual study behaviour. If room stock, access rules or sensor coverage have changed materially, downgrade confidence or withhold the forecast. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Helps the doctoral researcher plan around predictable pressure peaks instead of losing time searching after spaces fill.

1. Long-horizon planning, irregular schedules and focused work

Questions PHD-005-PHD-006

PHD-005

CORE | L4 - day-plan optimisation | R2 - integration-dependent

PhD student question: I have six hours on campus for thesis writing, data analysis and two short breaks. Can you plan a realistic sequence of spaces with independent backups?

Sensors and telemetry

Anonymous occupancy, room sound statistics, temperature, CO2 and illuminance; corridor and stair footfall; aggregate network, power and workstation-service status; lift status where step-free routing is required. No personal movement tracking. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Access and opening hours, room purposes and permitted activities, bookings, route graph, current workstation or software inventory, closure status and the student's preferred start time and accessibility constraints. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: verified permission, access, service, accessibility and timing for every candidate. Method: treat access, activity suitability, minimum session length and transfer time as hard constraints; forecast conditions for each interval; minimise unnecessary movement; place breaks in verified suitable areas and provide a fallback for each work block. Show trade-offs, an independent backup and uncertainty.

Answer boundary

The service must not prescribe productivity, reserve spaces or infer whether breaks were taken. A quiet room is not necessarily permitted for long study or data analysis. If any required service or route is unverified, return only the validated parts of the plan. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Turns an irregular doctoral workday into a feasible schedule without treating the building as a productivity monitor.

PHD-006

CORE | L3 - activity-specific recommendation | R2 - integration-dependent

PhD student question: Can one permitted workspace support both close reading and computational work today, or should I plan to use two different spaces?

Sensors and telemetry

Room-level sound, illuminance, temperature and CO2; anonymous occupancy; aggregate network performance and room electrical-fault status. For managed computer spaces, use authorised endpoint availability rather than environmental inference. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current room purpose, permitted activity, power provision, bookability, workstation or software availability, opening hours and student access. The system must not inspect documents, code, datasets or browser activity. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: translate each task into explicit constraints, compare available spaces on quietness, lighting, environmental condition, power and computing service, and recommend one shared location only if it meets both sets of requirements. Show primary option, backup, trade-offs and evidence time.

Answer boundary

Do not infer cognitive performance from sensor values or claim that a location will improve reading or coding. Network statistics are aggregate and cannot guarantee a particular device session. If task requirements are not supplied, ask for clarification. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Helps a doctoral student choose spaces based on the actual demands of different research activities.

1. Long-horizon planning, irregular schedules and focused work

Questions PHD-007-PHD-008

PHD-007

SITUATIONAL | L4 - recurrence diagnosis | R3 - research-grade or restricted

PhD student question: This room feels warmer and noisier than usual. Is today an isolated episode, or part of a recurring pattern worth reporting?

Sensors and telemetry

Room-level acoustic statistics, temperature, RH, CO_2 and anonymous occupancy; equipment or plug-load energy where available; HVAC state and outdoor weather; sensor health and maintenance events. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Verified room identity and use, scheduled events, maintenance activity, current fault records and the student's voluntary report time. Do not record speech or inspect research activity. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: time-aligned target, context and system streams covering baseline, event and recovery. Method: compare the current episode with matched periods, calculate duration and recurrence, align occupancy, weather, equipment load and HVAC changes, and rank supported explanations without asserting causality. Test confounders and alternatives; return supported, conflicting or not assessable.

Answer boundary

A single complaint or sensor spike cannot establish a fault. Sound level does not identify the source, and temperature cannot diagnose occupant health. If room coverage, baseline comparability or operational data are missing, provide a limited pattern summary and reporting route. If a critical source is missing, stale, conflicting or unauthorised, restrict disclosure, provide only verified facts, or return not assessable.

Why this is useful: Helps distinguish a short-lived disruption from a persistent problem that deserves a facilities report.

PHD-008

CORE | L3 - recurring schedule recommendation | R2 - integration-dependent

PhD student question: Each week, when am I most likely to get two quiet hours for difficult reading before my regular seminar?

Sensors and telemetry

Historical anonymous occupancy, doorway counts, room sound statistics, temperature, CO_2 and illuminance; corridor footfall around seminar changeovers. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

The seminar time and location supplied by the student, current student spaces, bookings, teaching timetable, opening hours, route graph and event schedule. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: search recurring pre-seminar windows, forecast demand and conditions from comparable weeks, account for class-change footfall and walking time, and rank several repeatable options with confidence. Show primary option, backup, trade-offs and evidence time.

Answer boundary

Do not retain the student's recurring schedule beyond the chosen service or infer attendance. A historical pattern may fail during special events or deadline peaks, so the answer must state exceptions and a refresh point. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Supports a sustainable weekly research routine around fixed academic commitments.

1. Long-horizon planning, irregular schedules and focused work

Questions PHD-009-PHD-010

PHD-009

CORE | L4 - commute-aware weekly optimisation | R2 - integration-dependent

PhD student question: I can travel to campus on only two days next week. Which two days would make best use of the journey for writing, supervision and authorised specialist computing?

Sensors and telemetry

Forecast occupancy, sound and environmental conditions in candidate spaces; aggregate network and managed-computing service status; corridor footfall and lift status for route feasibility. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Supervision and computing bookings voluntarily supplied by the student, opening hours, room access, bookability, current specialist-service inventory, planned maintenance, weather and public transport information from authoritative feeds. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: verified permission, access, service, accessibility and timing for every candidate. Method: treat fixed appointments and access as constraints, compare viable days, forecast space and service conditions, estimate transfer and travel buffers, and return two options with sensitivity to transport or service disruption. Show trade-offs, an independent backup and uncertainty.

Answer boundary

The service must not infer the student's home, finances or personal circumstances. Public transport and building conditions are forecasts, not guarantees. Do not recommend a day if specialist access or required service status is unverified. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Helps commuting and part-time doctoral researchers concentrate campus-dependent work into efficient visits.

PHD-010

ADVANCED | L4 - longitudinal change analysis | R3 - research-grade or restricted

PhD student question: Has the environment in my usual workspace changed since term began, and can you separate genuine change from altered sensing, room use or schedules?

Sensors and telemetry

Unchanged or versioned room-level temperature, RH, direct NDIR CO2, illuminance, acoustic statistics and anonymous occupancy; relevant BMS and outdoor data; sensor relocation, replacement, firmware, calibration and downtime history. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Verified room-use history, maintenance and refurbishment records, timetable or event context and any access changes. Use only the area identified by the student; do not analyse individual presence or work output. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: overlapping, equivalently placed streams above a declared coverage threshold. Method: identify any sensor/system or room-use changes first; segment the term into comparable periods; compare distributions, variability and recurrence after adjusting for occupancy and weather; separate measurement change from environmental change. Report effect size, coverage, uncertainty and sensitivity.

Answer boundary

A detected shift may result from sensor replacement, room-use change, HVAC work or schedule change. Do not label it improvement or deterioration without a defined criterion. If the pre/post measurement chain is not comparable, limit the answer to documented changes. If a critical source is missing, stale, conflicting or unauthorised, restrict disclosure, provide only verified facts, or return not assessable.

Why this is useful: Helps the doctoral researcher distinguish a genuine environmental change from altered sensing, building operation or room use before drawing a thesis or facilities conclusion.

2. Supervision, progress reviews and doctoral milestones

Questions PHD-011-PHD-012

PHD-011

CORE | L2 - confidential-room recommendation | R2 - integration-dependent

PhD student question: I need a confidential supervision meeting tomorrow. Which available room is officially suitable, quiet and step-free if I need it?

Sensors and telemetry

Current room-level acoustic statistics, anonymous occupancy, temperature and CO_2; corridor footfall for approach context. Acoustic sensing must retain numerical levels only, never speech. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current bookable-room catalogue, student entitlement, room enclosure/privacy features, capacity, accessibility, opening hours, bookings, route and closure status. A historic room label or a low sound reading cannot establish confidentiality. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: authoritative lookup plus calculation or bounded comparison. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: filter by verified permission, enclosure, capacity, accessibility and availability; use measured sound and nearby footfall only as supporting evidence; rank primary and backup rooms and provide booking instructions. Show primary option, backup, trade-offs and evidence time.

Answer boundary

Quietness and an empty-room estimate do not establish confidentiality. Use only rooms officially designated and bookable for the meeting, disclose no attendee identity, and treat accessibility as a hard requirement. If room purpose, booking, access or privacy status is unverified, do not recommend it.

Why this is useful: Enables a doctoral researcher to arrange sensitive supervision without mistaking quietness or apparent vacancy for verified confidentiality.

PHD-012

CORE | L3 - hybrid-supervision readiness | R2 - integration-dependent

PhD student question: My supervisor will join remotely. Which bookable room is most likely to support a reliable call without disturbing other students?

Sensors and telemetry

Room sound statistics, anonymous occupancy, temperature and CO_2; authorised aggregate wireless/network status, AV endpoint status and electrical-fault status. No inspection of call content, applications or device identifiers. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current bookable-room inventory, permitted call use, bookings, capacity, power and verified conferencing equipment; student access and route data; network incident feed. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: current official record or target-room observation with valid spatial coverage. Method: confirm official service availability, compare current and recent room conditions, filter for enclosed or call-permitted spaces, and rank options with a pre-call recheck and fallback. Return source and observed-at time; do not infer beyond status.

Answer boundary

Aggregate network health cannot guarantee a particular laptop or platform. Quietness does not prove confidentiality. Do not reserve or join the call, and do not expose other users' bookings. If service status is unknown, label the recommendation conditional. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Reduces avoidable disruption during hybrid supervision and directs the PhD student to a space where calls are permitted.

2. Supervision, progress reviews and doctoral milestones

Questions PHD-013-PHD-014

PHD-013

CORE | L3 - review-room forecast | R2 - integration-dependent

PhD student question: Which room should I request for my annual progress review if I need four seats, a working display, low background noise and step-free access?

Sensors and telemetry

Room-level acoustic statistics, temperature, CO_2 and anonymous occupancy; official lift and door status near the event time; authorised AV and network service status. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current room capacity, furniture, bookability, display/conferencing equipment, access permissions, accessible route, review duration and bookings. Current facts must come from operational records, not historic labels. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified future status plus comparable quality-controlled history. Method: apply hard requirements for capacity, accessibility, display and permission; forecast nearby demand and room conditions for the review period; rank candidates and explain any trade-off. Report horizon, coverage, prediction interval and recheck trigger.

Answer boundary

The building cannot determine whether a room meets academic review policy unless that policy is supplied by the institution. Lift and network status can change. Do not reveal panel identities or schedules beyond the authorised booking context. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Supports a formal milestone meeting with fewer environmental, access and equipment risks.

PHD-014

CORE | L4 - milestone-day itinerary | R2 - integration-dependent

PhD student question: On my progress-review day, I need an hour to rehearse, a quiet place to wait and the review itself. Can you plan the sequence with realistic transition time?

Sensors and telemetry

Anonymous occupancy, room sound, temperature, CO_2 and illuminance; corridor footfall; aggregate network/AV status and lift status where required. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Review booking and rehearsal requirements supplied by the student, current room purpose and permitted use, bookability, opening hours, route graph, access and accessibility data. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: verified permission, access, service, accessibility and timing for every candidate. Method: allocate distinct spaces for rehearsal, waiting and review; enforce booking, privacy, accessibility and transfer-time constraints; forecast conditions and crowding; provide fallback rooms and recheck times. Show trade-offs, an independent backup and uncertainty.

Answer boundary

The system must not infer the review outcome, monitor rehearsal content or expose panel information. A waiting space may be quiet without being private. If any essential room or route status is unverified, offer only the confirmed part of the itinerary. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Reduces uncertainty on a high-pressure milestone day while preserving confidentiality and role permissions.

2. Supervision, progress reviews and doctoral milestones

Questions PHD-015-PHD-016

PHD-015

CORE | L2 - pre-event service check | R2 - integration-dependent

PhD student question: My milestone presentation begins in 30 minutes. What do official room and service feeds and current room sensors confirm, and what must I still test myself?

Sensors and telemetry

Room temperature, RH and direct NDIR CO2 for current conditions; authoritative AV endpoint heartbeat, display or controller status, aggregate access-point/service health and room-circuit fault state. Environmental sensors cannot prove that the display, adapter, student device or presentation file will work. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: IT uses authorised timestamped aggregate heartbeat/status only; no traffic, research content or identity; equipment state comes from authorised service logs, not room conditions or power alone.

Authoritative supporting data

The student's authenticated booking, current room/equipment inventory, AV and network service status, planned outages, access and presentation time. The assistant must not inspect the presentation, files, device configuration or private calendar details beyond what is shared. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: authoritative lookup plus current observation. Minimum evidence: a valid booking and current official AV, network, access and closure status, supplemented by correctly located room sensors. Method: confirm the room and start time; report service states and current conditions with timestamps; separate remotely verifiable facts from a short manual checklist for display, microphone, camera and content playback. Output: confirmed, unconfirmed and manual-test items.

Answer boundary

A green service heartbeat is not proof that the student's laptop, adapter, file or platform will work. Do not inspect files, credentials or traffic. Any endpoint lacking current telemetry must be marked unverified and manually tested; safety-critical or access issues go to the responsible service. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Gives the PhD student a precise pre-presentation checklist while preserving the need for a final hands-on equipment test.

PHD-016

SITUATIONAL | L4 - inclusive panel-room comparison | R2 - integration-dependent

PhD student question: Two rooms are available for my progress review. Which is better once accessibility, confidentiality, hybrid participation and likely room conditions are considered together?

Sensors and telemetry

Room sound statistics, temperature, CO2, illuminance and anonymous occupancy; official lift/door status; aggregate network and conferencing endpoint status. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current capacity, layout, privacy features, accessible-route information, booking and access rules, hybrid equipment, panel requirements voluntarily supplied and room availability. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: overlapping, equivalently placed streams above a declared coverage threshold. Method: treat accessibility, permission, capacity and required equipment as hard constraints; compare privacy evidence, forecast conditions and service reliability; show the weighting and explain why one room is preferred. Report effect size, coverage, uncertainty and sensitivity.

Answer boundary

Do not infer disability or panel preferences. Acoustic levels do not certify privacy, and no option should be ranked accessible if any required route segment is unverified. If both rooms fail a hard requirement, recommend neither. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Makes a consequential room choice transparent and inclusive rather than relying on nominal capacity alone.

2. Supervision, progress reviews and doctoral milestones

Questions PHD-017-PHD-018

PHD-017

SITUATIONAL | L4 - recurring disruption assessment | R3 - research-grade or restricted

PhD student question: Our regular supervision slot is often disrupted by nearby noise or crowding. Does the evidence support that pattern, and would another time be more dependable?

Sensors and telemetry

Local room and nearby corridor acoustic statistics, anonymous occupancy and footfall, door state where appropriate, plus temperature and CO_2 for candidate alternatives. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

The recurring slot and room voluntarily provided, timetable, bookings, event schedule, current room use and access. Do not use supervisor or student calendars beyond consented inputs. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: time-aligned target, context and system streams covering baseline, event and recovery. Method: quantify disturbance indicators across comparable weeks, test whether peaks align with scheduled changeovers, forecast alternative slots and rank those that preserve room and participant constraints. Test confounders and alternatives; return supported, conflicting or not assessable.

Answer boundary

Sound and footfall data indicate activity, not who caused it or whether speech was intelligible. Sparse weeks or changed schedules may make recurrence uncertain. Do not automatically change a meeting or disclose other bookings. If a critical source is missing, stale, conflicting or unauthorised, restrict disclosure, provide only verified facts, or return not assessable.

Why this is useful: Helps a supervisory pair choose a more dependable slot using aggregate evidence rather than anecdote.

PHD-018

CORE | L3 - visitor route and arrival plan | R2 - integration-dependent

PhD student question: An external reviewer is coming to Abacws for my progress meeting. What verified entrance, step-free route and waiting arrangement should I send them?

Sensors and telemetry

Official lift and door state, corridor footfall and broad waiting-area occupancy; environmental sensing in public waiting areas if a comfort recommendation is requested. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; IT uses authorised timestamped aggregate heartbeat/status only; no traffic, research content or identity.

Authoritative supporting data

Current public entrances, visitor access procedure, meeting room, accessible route graph, reception information, planned closures, event time and any accessibility requirements the visitor chooses to disclose. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: current official record or target-room observation with valid spatial coverage. Method: select an authorised entrance, compute a valid route, estimate travel time, identify a verified public waiting point and provide a fallback if a lift or route is unavailable. Return source and observed-at time; do not infer beyond status.

Answer boundary

Do not expose restricted areas, access credentials, other visitors or live security detail. The system cannot guarantee admission or lift availability; official visitor instructions and on-site signage take precedence. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Allows the PhD student to give a reviewer clear, accessible and current arrival information.

2. Supervision, progress reviews and doctoral milestones

Questions PHD-019-PHD-020

PHD-019

SITUATIONAL | L3 - document-review room recommendation | R2 - integration-dependent

PhD student question: I need a two-hour chapter-review meeting using printed drafts and a shared display. Which bookable room has suitable table space, lighting and privacy?

Sensors and telemetry

Task-area illuminance, room sound statistics, temperature, CO2 and anonymous occupancy; authorised display and power status. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current room furniture and capacity, bookability, permitted use, display/power inventory, access, opening hours and bookings. The system must not inspect drafts or meeting content. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: filter for verified table, display, capacity and access requirements; compare lighting, noise and environmental condition; rank options and state which features are confirmed versus only observed. Show primary option, backup, trade-offs and evidence time.

Answer boundary

Illuminance at a sensor point may not represent every seat, and a quiet room is not necessarily confidential. Do not infer document sensitivity. If furniture or equipment inventory is stale, require confirmation before recommendation. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Supports detailed feedback sessions where table layout, visual conditions and low disruption matter.

PHD-020

SITUATIONAL | L3 - post-review transition recommendation | R2 - integration-dependent

PhD student question: After my review, I have 90 minutes before teaching. Where can I work quietly and still reach the teaching room with a sensible buffer?

Sensors and telemetry

Current and near-term occupancy, room sound statistics, temperature, CO2 and corridor footfall; official lift status if required. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Teaching time and location supplied by the student, current student spaces, access, bookings, opening hours and route graph. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: calculate the usable work interval after including a safe travel buffer, compare nearby eligible spaces, forecast short-term conditions and return a primary and backup location with a departure time. Show primary option, backup, trade-offs and evidence time.

Answer boundary

Do not infer review results or teaching attendance. A short-term forecast cannot guarantee a seat. If the teaching room or route changes, the answer must be refreshed rather than reused. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Preserves a useful work interval while protecting a fixed teaching-assistance commitment.

3. Experiments, data collection, specialist computing and equipment

Questions PHD-021-PHD-022

PHD-021

CORE | L3 - stable-condition forecast | R3 - research-grade or restricted

PhD student question: I have an authorised environment-sensitive experiment this week. When is the assigned project space likely to have the most stable temperature, humidity, particulate level and background sound?

Sensors and telemetry

Experiment-appropriate, correctly located temperature and RH; PM2.5/PM10 only where particle stability matters and the instrument is validated; numerical acoustic statistics only where vibration or sound is relevant; anonymous occupancy; equipment load, extraction/HVAC state and outdoor reference. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: instrument range, response time and sampling must match the approved protocol; sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech.

Authoritative supporting data

Current room purpose, student permission, equipment booking, induction, approved protocol/risk assessment, planned activities, cleaning and maintenance. The May 2021 room label does not prove present experimental use or authorisation. The official safety source prevails; retain source, issue time, validity window and permission.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: protocol-matched room measurements, authorised window and comparable quality-controlled history. Method: translate the approved protocol into stability limits and duration; calculate variance/event frequency; adjust for occupancy, equipment, cleaning, weather and HVAC; rank authorised windows. Report prediction intervals and a recheck trigger.

Answer boundary

This is an environmental-stability forecast, not experimental validation or permission to proceed. Building sensors cannot certify instrument suitability, protocol compliance or research safety. If the assigned space, coverage, calibration, activity schedule or authorisation is unverified, return the verified facts and mark the forecast not assessable.

Why this is useful: Supports defensible scheduling of environment-sensitive doctoral work without confusing building telemetry with experimental validation.

PHD-022

ADVANCED | L4 - event-source diagnosis | R3 - research-grade or restricted

PhD student question: During yesterday's data collection, PM and VOC readings increased. Did the timing align with equipment use, occupancy, an open door, extraction changes, cleaning or outdoor pollution?

Sensors and telemetry

Validated room PM2.5/PM10 and indicative TVOC, temperature and RH; equipment power or authorised event log; anonymous occupancy; door state; extraction airflow/fan state; cleaning/maintenance events; outdoor PM and weather. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: instrument range, response time and sampling must match the approved protocol; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Authorised experiment schedule and equipment log voluntarily linked to the query, room maintenance and cleaning records, extraction-system events and current room use. No inspection of research content. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: quality-flagged room PM/TVOC, equipment/activity, door, extraction, cleaning and outdoor streams aligned through baseline, event and recovery. Method: reconstruct onset, peak and decay; test time lags and competing explanations. Report observed change, supported associations, conflicts and unobserved causes; never identify a compound or assert causality.

Answer boundary

Indicative TVOC cannot identify a chemical or establish exposure safety, and temporal alignment does not prove causation. Certified systems and the approved risk assessment take precedence. Conflicting clocks, gaps or missing source logs require a qualified or not-assessable diagnosis. If a critical source is missing, stale, conflicting or unauthorised, restrict disclosure, provide only verified facts, or return not assessable.

Why this is useful: Supports a reproducible investigation of an environmental or data-quality event while keeping chemical identification and safety judgement with appropriate methods and people.

3. Experiments, data collection, specialist computing and equipment

Questions PHD-023-PHD-024

PHD-023

CORE | L1 - authorised service lookup | R2 - integration-dependent

PhD student question: Is an authorised specialist-computing space available now with the exact software, hardware and licence entitlement I need?

Sensors and telemetry

Optional anonymous occupancy and room environmental status; authorised aggregate workstation availability and service health. Environmental sensors cannot verify installed software or user entitlement. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: instrument range, response time and sampling must match the approved protocol; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current specialist-computing catalogue, student entitlement, induction or licence conditions, workstation specification, software/version availability, booking/opening hours, maintenance window and live service status. Technical availability never implies permission. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: authoritative lookup and/or current observation. Minimum evidence: current official record or target-room observation with valid spatial coverage. Method: match the student-stated software and hardware requirements against authorised current inventory, verify access and service status, and present eligible options with official instructions. Return source and observed-at time; do not infer beyond status.

Answer boundary

A room that appears empty is not necessarily available, and the system cannot grant software, licence, account or equipment entitlement. Do not inspect files, jobs or research content. If the authoritative inventory, access or live service state is unavailable, provide the service desk or booking route rather than infer availability.

Why this is useful: Prevents wasted travel to a specialist space that cannot support the required research software or access level.

PHD-024

CORE | L3 - long-compute session recommendation | R2 - integration-dependent

PhD student question: I need to analyse data on campus for most of the day. Which permitted workspace has dependable power and network service and is least likely to become crowded or uncomfortable?

Sensors and telemetry

Room temperature, CO_2, illuminance and sound statistics; anonymous occupancy; aggregate network status and electrical fault or circuit status. For managed workstations, use authorised endpoint service data. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: instrument range, response time and sampling must match the approved protocol; sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech.

Authoritative supporting data

Current access, permitted use, power and workstation provision, opening hours, bookings, network incidents, planned outages and software or service requirements stated by the student. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: filter on verified services and access, forecast occupancy and environmental conditions across the requested duration, assess service incident history and rank a primary and backup option with recheck points. Show primary option, backup, trade-offs and evidence time.

Answer boundary

The service cannot guarantee a specific socket, laptop battery, device connection or computation outcome. Do not inspect data or monitor the student's session. If power/network telemetry is only building-wide, state that room-level reliability is unverified. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Supports long computational work without conflating general building conditions with managed computing-service guarantees.

3. Experiments, data collection, specialist computing and equipment

Questions PHD-025-PHD-026

PHD-025

SITUATIONAL | L4 - equipment-environment diagnosis | R3 - research-grade or restricted

PhD student question: Could measured room conditions plausibly be contributing to my equipment instability, and what evidence would be required before I treat that as more than a hypothesis?

Sensors and telemetry

Protocol-specific temperature, RH, vibration, particulate or electromagnetic-interference measurement only where physically relevant; equipment power/current and authorised diagnostic status; HVAC state; door and occupancy context; calibrated time synchronisation between equipment and building data. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: instrument range, response time and sampling must match the approved protocol; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Equipment specification and operating limits, authorised maintenance log, experiment timestamps, room service relationships and risk assessment. Do not access research outputs beyond the minimum diagnostic metadata approved for the query. The official safety source prevails; retain source, issue time, validity window and permission.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: time-aligned equipment-status metadata and protocol-relevant room/power measurements covering stable and unstable periods. Method: compare with documented limits and matched baselines; test lagged and competing explanations. Return supported, conflicting or not assessable with coverage.

Answer boundary

Building telemetry cannot replace manufacturer diagnostics, instrument calibration or competent inspection. Association does not prove that the room caused the instability, and research content must remain unseen. If equipment logs, environmental coverage or timing are inadequate, state the plausible hypotheses and return the diagnosis not assessable.

Why this is useful: Helps decide whether the next investigation should focus on equipment, room conditions or both without exposing research content or overstating causality.

PHD-026

ADVANCED | L4 - deployment-site comparison | R3 - research-grade or restricted

PhD student question: I have approval to place a temporary research sensor at one of two sites. Which site offers more representative exposure, stable power and network, low interference and safe maintenance access?

Sensors and telemetry

Site-specific power quality or circuit availability, aggregate radio/network service measurements, temperature/RH and relevant environmental gradients, footfall and door context, and co-location with a traceable reference sensor. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: instrument range, response time and sampling must match the approved protocol; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Written deployment approval, current room/corridor purpose, mounting and electrical restrictions, network owner approval, fire/accessibility constraints, maintenance access, removal date and research-ethics/data-protection approval where required. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: overlapping, equivalently placed streams above a declared coverage threshold. Method: define the measurement objective, compare interference, representativeness, service reliability and maintenance access, identify bias introduced by each site, and recommend a location only within the authorised options. Report effect size, coverage, uncertainty and sensitivity.

Answer boundary

The system cannot grant deployment permission or certify electrical, fire, accessibility, network or ethics compliance. A technically convenient location may be scientifically unrepresentative. If the measurand, mounting geometry or authorisation is unclear, ask for clarification and do not rank. If a critical source is missing, stale, conflicting or unauthorised, restrict disclosure, provide only verified facts, or return not assessable.

Why this is useful: Improves the scientific representativeness and maintainability of a temporary deployment while keeping permission and ethics outside the chatbot.

3. Experiments, data collection, specialist computing and equipment

Questions PHD-027-PHD-028

PHD-027

SITUATIONAL | L3 - low-interference collection forecast | R3 - research-grade or restricted

PhD student question: When can I collect room-level environmental data with minimal disruption from class changeovers and unusual events while still observing normal building use?

Sensors and telemetry

Corridor footfall, doorway counts, anonymous room occupancy, door state, room environmental sensors and event-quality flags. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; access/lift/door/closure uses authoritative timestamped events.

Authoritative supporting data

Timetable, bookings, planned events, maintenance and current room access. The research protocol must define whether normal occupancy or an unoccupied baseline is required. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified future status plus comparable quality-controlled history. Method: classify candidate periods by routine use, changeover intensity and unusual events; estimate expected occupancy and disturbance; provide several windows and describe the population or activity each would represent. Report horizon, coverage, prediction interval and recheck trigger.

Answer boundary

A low-interference period may not be representative of the research question. Do not manipulate access or observe identifiable individuals. Research ethics, consent and room permission remain external requirements. If a critical source is missing, stale, conflicting or unauthorised, restrict disclosure, provide only verified facts, or return not assessable.

Why this is useful: Helps the doctoral researcher choose a sampling window that is both practical and scientifically defensible.

PHD-028

CORE | L4 - dataset fitness assessment | R3 - research-grade or restricted

PhD student question: Which Abacws sensor streams are methodologically suitable for my thesis analysis, and what placement, calibration, drift, missing-data and selection biases must I account for?

Sensors and telemetry

Full sensor registry with observed property, units, exact deployment zone, mounting height/orientation, make/model, stated accuracy, calibration/commissioning, sampling and aggregation, firmware, clock quality, uptime, missingness, drift checks, relocation/replacement and maintenance history. Include BMS point quality where used. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: BMS uses native/event points with a 1-min historian and verified asset-zone command/feedback.

Authoritative supporting data

Ontology/time-series mapping, data dictionary, deployment and relocation history, calibration and maintenance records, current room/HVAC model, access licence, data owner, governance conditions and known incidents. Proposed sensors must not be described as available datasets. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: registry, deployment, calibration, clock, maintenance and permission metadata for each candidate stream. Method: map streams to the thesis variable and required scale; quantify completeness, drift, discontinuities, spatial representativeness, selection bias and cross-source agreement. Classify suitable, conditionally suitable or unsuitable with reproducible reasons.

Answer boundary

Dataset availability is not methodological suitability or permission. Corridor streams cannot silently support room-level claims, and an undocumented processing chain cannot support a reproducible thesis result. If provenance, placement, calibration, sampling, coverage or governance is inadequate, identify the gap and mark the proposed use not assessable.

Why this is useful: Helps the doctoral researcher decide whether a stream is fit for the proposed thesis method before investing in analysis.

3. Experiments, data collection, specialist computing and equipment

Questions PHD-029-PHD-030

PHD-029

CORE | L4 - sensor-quality diagnosis | R3 - research-grade or restricted

PhD student question: One sensor stream is drifting away from comparable nearby measurements and contains gaps. Is that more consistent with a genuine local difference or a sensor-quality problem?

Sensors and telemetry

The target stream, physically comparable reference streams, raw quality flags, calibration/commissioning and maintenance records, clock status, power/communications health, location changes and relevant local drivers. Nearby sensors are valid comparators only if they observe comparable conditions. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; BMS uses native/event points with a 1-min historian and verified asset-zone command/feedback.

Authoritative supporting data

Sensor model and accuracy, installation and relocation records, maintenance tickets, firmware changes, room/HVAC relationships and any known local source activity. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: diagnosis and evidence-quality audit. Minimum evidence: the target stream, physically comparable references, deployment history and an overlap period above a declared coverage threshold. Method: align clocks and units; inspect gaps, step changes, flat-lines, calibration or firmware events and divergence conditional on occupancy, weather and local sources; compare alternative explanations; classify the evidence as local difference, sensor problem, conflicting or not assessable. Output: diagnostics, coverage and recommended validation.

Answer boundary

Divergence from nearby sensors is not automatically drift; local airflow, solar gain or occupancy may be real. Conversely, agreement does not prove accuracy. If no independent reference or maintenance evidence exists, return competing explanations with uncertainty. If a critical source is missing, stale, conflicting or unauthorised, restrict disclosure, provide only verified facts, or return not assessable.

Why this is useful: Prevents a local environmental effect from being discarded as bad data and prevents a faulty sensor from driving a false research conclusion.

PHD-030

CORE | L3 - equipment-movement plan | R2 - integration-dependent

PhD student question: I need to move approved equipment from the entrance to a project space. Which authorised route and time would minimise stairs, congestion and handling risk?

Sensors and telemetry

Official lift and door status, corridor/stair footfall and broad route occupancy; optional environmental status for equipment-sensitive transport. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: instrument range, response time and sampling must match the approved protocol; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current authorised destination, equipment movement approval, size/weight and handling constraints, accessible/service route graph, door/lift dimensions, access permissions, planned closures and local safety procedure. The official safety source prevails; retain source, issue time, validity window and permission.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: current official policy, alarm or approval; sensor streams are supplementary. Method: remove routes that fail dimensions, access or step-free constraints; forecast congestion; calculate travel and handling time; provide a primary route, fallback and recheck point. Report source and issue time; never infer clearance or emergency action.

Answer boundary

The conversational service cannot approve manual handling, hazardous transport or use of service routes. Official risk assessment, competent assistance and security instructions take precedence. If dimensions or route status are unverified, do not recommend movement. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Reduces avoidable handling and access problems when doctoral work depends on moving authorised equipment.

4. Collaboration, seminars, teaching assistance and research events

Questions PHD-031-PHD-032

PHD-031

CORE | L3 - research-group room recommendation | R2 - integration-dependent

PhD student question: Our research group needs a 90-minute meeting for eight people. Which available room supports discussion, a shared display, step-free access and acceptable room conditions?

Sensors and telemetry

Room CO2, temperature, sound statistics and anonymous occupancy; official lift status; aggregate display, network and power status. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current capacity, furniture, bookability, permitted group use, display/conferencing equipment, student access, accessibility, bookings and meeting time. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: filter on capacity, access, equipment and accessibility; forecast occupancy and room conditions; rank a primary and backup room and show any trade-offs. Show primary option, backup, trade-offs and evidence time.

Answer boundary

Do not expose participant identities or other booking details. CO2 is a ventilation indicator, not complete air quality, and a room meeting nominal criteria is not guaranteed comfortable for everyone. If a hard requirement is unverified, exclude the room. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Supports efficient research collaboration without treating nominal capacity as the only suitability criterion.

PHD-032

CORE | L3 - recurring reading-group schedule | R2 - integration-dependent

PhD student question: What recurring time and bookable room would give our doctoral reading group the best chance of a quiet, dependable weekly meeting?

Sensors and telemetry

Historical anonymous occupancy, corridor footfall, room sound, temperature and CO2; aggregate AV/network status if hybrid attendance is needed. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Group constraints voluntarily supplied, current room access and permitted use, recurring bookings, timetable, event schedule and opening hours. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: identify feasible weekly slots, model demand and disturbance, filter by access and booking, rank room-time pairs and state exceptions such as deadline or event weeks. Show primary option, backup, trade-offs and evidence time.

Answer boundary

Do not infer members' calendars or attendance. Historical quiet does not guarantee speech privacy or future availability. Shared schedule data requires consent and should be retained only for the service purpose. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Helps a research group establish a stable routine with fewer repeated booking and disturbance problems.

4. Collaboration, seminars, teaching assistance and research events

Questions PHD-033-PHD-034

PHD-033

CORE | L2 - teaching-assistance readiness | R2 - integration-dependent

PhD student question: I am supporting a teaching lab in an hour. Is the room officially open, are the main teaching systems available, and what do current room measurements show for the scheduled class size?

Sensors and telemetry

Room temperature, CO_2 and RH; anonymous occupancy/doorway count; authorised AV, network and managed-workstation status; HVAC state where available. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; BMS uses native/event points with a 1-min historian and verified asset-zone command/feedback.

Authoritative supporting data

Teaching timetable, current room allocation, scheduled class size, official opening/access status, AV/IT service status, maintenance and responsible teaching-team instructions. The PhD student is not authorised to declare the room operational or safe. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: authoritative lookup plus calculation or bounded comparison. Minimum evidence: current official record or target-room observation with valid spatial coverage. Method: verify room and service state, compare current occupancy and environmental trend with the scheduled load, identify active incidents and provide the official escalation route. Return source and observed-at time; do not infer beyond status.

Answer boundary

The assistant may report official readiness and current environmental observations, but it cannot open the room, authorise teaching, change systems or certify that every workstation works. It must not expose student identities or attendance. If the official room or service feed is unavailable, direct the user to teaching support and label the readiness check incomplete.

Why this is useful: Allows a PhD teaching assistant to detect practical room or service problems before students arrive.

PHD-034

SITUATIONAL | L3 - student-support space recommendation | R2 - integration-dependent

PhD student question: I am holding a short student drop-in session. Which permitted space is easy to find, allows conversation and is unlikely to become overcrowded?

Sensors and telemetry

Anonymous occupancy, corridor footfall, room sound, temperature and CO_2; official lift status if accessibility is relevant. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Teaching-assistant permission, current student-facing room inventory, bookability, permitted activity, capacity, accessibility, route and opening hours. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: filter for authorised student-facing use and capacity, forecast footfall and room demand, compare current conditions and rank options with clear directions. Show primary option, backup, trade-offs and evidence time.

Answer boundary

Do not infer which students will attend or publish sensitive attendance information. A public or open area may not suit confidential discussion. If the intended support is private, require an appropriately verified room. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Supports approachable student help while respecting capacity, privacy and building-use rules.

4. Collaboration, seminars, teaching assistance and research events

Questions PHD-035-PHD-036

PHD-035

CORE | L3 - collaborative coding-session recommendation | R2 - integration-dependent

PhD student question: Four of us need a three-hour coding session with power, reliable network access and permission to talk. Where should we work?

Sensors and telemetry

Anonymous occupancy, room sound, temperature and CO2; aggregate network status and electrical-fault status; authorised workstation service data if required. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current room purpose, group-work permission, capacity, bookability, power provision, access, opening hours and network incidents. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: apply capacity, talk-permission, power and access as hard constraints; forecast occupancy and conditions; rank a primary and backup space and state which services are officially verified. Show primary option, backup, trade-offs and evidence time.

Answer boundary

The system cannot guarantee sockets, bandwidth, licences or a particular device connection. Do not inspect code repositories or network traffic. If group discussion is not permitted, exclude the space even if it is quiet and empty. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Helps a doctoral project team find a technically and socially suitable collaboration space.

PHD-036

SITUATIONAL | L4 - visiting-researcher itinerary | R2 - integration-dependent

PhD student question: A visiting researcher will spend half a day with us. Can you plan an accessible sequence for arrival, a meeting, a seminar and a quiet room for an online call?

Sensors and telemetry

Official lift/door state, corridor footfall, room occupancy and sound, temperature and CO2; aggregate network/AV status. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Visitor procedure, permitted rooms, bookings, seminar schedule, public and restricted routes, accessibility requirements voluntarily supplied, reception and opening hours. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: verified permission, access, service, accessibility and timing for every candidate. Method: sequence authorised spaces with adequate transfer time, filter for call-permitted and accessible rooms, forecast crowding and conditions, and provide fallback routes and rooms. Show trade-offs, an independent backup and uncertainty.

Answer boundary

Do not expose restricted areas, access credentials or other occupants. The system cannot authorise visitor access or guarantee a confidential call. Official host and security procedures take precedence. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Helps a PhD host provide a coherent visit without improvising around access, hybrid and environmental constraints.

4. Collaboration, seminars, teaching assistance and research events

Questions PHD-037-PHD-038

PHD-037

SITUATIONAL | L4 - research-showcase forecast | R3 - research-grade or restricted

PhD student question: We are planning a poster-and-demonstration event. Where are crowding, overheating or ventilation problems most likely, and which timing or layout changes are supported by evidence?

Sensors and telemetry

Anonymous doorway counts and zone occupancy, corridor footfall, correctly located temperature and direct NDIR CO2 in candidate event zones, numerical sound statistics, supply-airflow/BMS state for the verified served zone, and optional equipment-power load. No camera identification or raw audio. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Approved event booking, current permitted use/capacity, expected attendance, furniture/layout approval, accessibility plan, AV/electrical provision, fire and stewarding requirements, maintenance/closures and responsible organiser decisions. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: authorised attendance/layout scenarios, room capacity, doorway throughput, anonymous counts, distributed room sensing and HVAC capability. Method: forecast crowding and CO2/temperature trajectories; compare timing, entry, break and layout mitigations. Report uncertainty and leave approval to the authorised organiser and safety roles.

Answer boundary

This is a planning forecast, not a crowd-control or safety approval. Use aggregate counts only, disclose no identities and do not recommend exceeding verified capacity or blocking accessible and emergency routes. If layout, capacity, booking, ventilation-zone or event assumptions are unverified, provide scenario ranges rather than a definitive plan.

Why this is useful: Helps a student organiser anticipate event pressure points and compare mitigations before committing to a room layout or schedule.

PHD-038

CORE | L3 - seminar-plus-work recommendation | R2 - integration-dependent

PhD student question: I am attending a seminar and want to write for two hours immediately afterwards. Which nearby permitted space is most likely to remain suitable after the audience leaves?

Sensors and telemetry

Current and historical room/nearby occupancy, corridor footfall, sound, temperature and CO2; aggregate network and power status. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Seminar time and location supplied by the student, current student spaces, bookings, access, opening hours and route graph. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: model audience dispersal and nearby demand after the seminar, filter eligible spaces, compare environmental recovery and noise, and rank a primary and backup option. Show primary option, backup, trade-offs and evidence time.

Answer boundary

The service cannot identify attendees or guarantee that a nearby seat remains free. Seminar overruns and unscheduled activity reduce confidence. If only corridor sensing is available for a candidate room, label room condition unknown. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Converts a seminar visit into a productive research block without unnecessary movement.

4. Collaboration, seminars, teaching assistance and research events

Questions PHD-039-PHD-040

PHD-039

ADVANCED | L4 - recurring seminar-space comparison | R3 - research-grade or restricted

PhD student question: Across this term, which seminar spaces have hosted recurring research events with the fewest crowding, noise, ventilation and service problems?

Sensors and telemetry

Room CO2, temperature, sound, anonymous occupancy and corridor footfall; aggregate AV/network incident data; sensor and service coverage history. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Verified seminar bookings and room uses, capacity, event attendance estimates where authorised, service tickets and room configuration changes. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: overlapping, equivalently placed streams above a declared coverage threshold. Method: define event-level metrics, adjust for attendance, duration and weather, account for missing sensing and service coverage, and compare spaces with uncertainty and event counts. Report effect size, coverage, uncertainty and sensitivity.

Answer boundary

The result cannot rank speakers, organisers or event quality, and spaces without equivalent evidence must not be penalised. Service tickets may be incomplete or differently recorded. Do not treat correlation as proof of room design failure. If a critical source is missing, stale, conflicting or unauthorised, restrict disclosure, provide only verified facts, or return not assessable.

Why this is useful: Provides evidence for choosing future doctoral seminar locations and targeting recurring operational problems.

PHD-040

CORE | L3 - hybrid-event readiness | R2 - integration-dependent

PhD student question: Our research seminar includes remote speakers. What do the official booking and service feeds and current room sensors confirm, and what still needs a manual test?

Sensors and telemetry

Current room temperature, RH and direct NDIR CO2; authoritative display/controller, conferencing endpoint and audio-system heartbeat where exposed; aggregate network service health and circuit fault status. Sound sensors are not microphone tests and must not record speech. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; IT uses authorised timestamped aggregate heartbeat/status only; no traffic, research content or identity.

Authoritative supporting data

Event booking, current room and AV inventory, authorised remote-speaker platform, network/AV incident feed, access/opening hours, responsible organiser and technical-support contacts. The assistant does not join, record or inspect the seminar. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: current official record or target-room observation with valid spatial coverage. Method: verify the booking and official system states, report room conditions, identify stale or missing endpoints, and return a pre-opening checklist with manual tests and escalation contacts. Do not infer audio quality from an acoustic sensor. Return source and observed-at time; do not infer beyond status.

Answer boundary

The system cannot guarantee device compatibility, platform performance or audio quality and must not record or analyse speech. Missing endpoint telemetry is unknown, not healthy. The responsible organiser or technical support must complete the manual sound and presentation test. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Reduces avoidable failure at a research seminar while clearly separating automated checks from the manual technical rehearsal.

5. Presentations, conference preparation and viva rehearsal

Questions PHD-041-PHD-042

PHD-041

CORE | L3 - rehearsal-room recommendation | R2 - integration-dependent

PhD student question: Where can I rehearse a 30-minute research talk using a display and standing space without disturbing people in quiet-study areas?

Sensors and telemetry

Room sound statistics, anonymous occupancy, temperature and CO_2; authorised display and power status. No recording of the talk by building sensors. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current room purpose, permitted rehearsal use, capacity/layout, bookability, access, display inventory, opening hours and bookings. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: filter for authorised rehearsal activity and equipment, compare current and forecast conditions, and rank a primary and backup room with booking guidance. Show primary option, backup, trade-offs and evidence time.

Answer boundary

A low sound level does not mean speaking is permitted, and room sensors cannot assess presentation quality. Do not record or transcribe rehearsal content. If standing space or equipment inventory is unverified, require a manual check. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Gives the PhD student a suitable place to practise delivery without disrupting quiet-study areas.

PHD-042

SITUATIONAL | L3 - recording-space recommendation | R3 - research-grade or restricted

PhD student question: I want to record a practice presentation on my own device. Which authorised room is likely to be quiet enough and has stable lighting, power and network service?

Sensors and telemetry

Numerical room sound statistics, task/face-area illuminance where available, temperature, anonymous occupancy and aggregate network status. The building must not capture raw audio or video. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Room permission for self-recording, bookability, privacy features, lighting controls, access, opening hours and network incidents. Any recording consent or data policy remains external. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: filter to rooms where self-recording is permitted, compare current sound and lighting stability, check official network state and rank options with a short pre-recording recheck. Show primary option, backup, trade-offs and evidence time.

Answer boundary

The building cannot grant recording permission or guarantee speech privacy. Use only a room whose booking terms permit recording, avoid capturing passers-by or confidential material, and store audio/video only on the student-controlled device under applicable policy. If room permission or privacy cannot be verified, recommend rehearsal without recording or another authorised option.

Why this is useful: Supports realistic rehearsal and self-recording while protecting privacy and respecting room-use rules.

5. Presentations, conference preparation and viva rehearsal

Questions PHD-043-PHD-044

PHD-043

CORE | L3 - mock-viva room recommendation | R2 - integration-dependent

PhD student question: We need a room for a two-hour mock viva with four people. Which option is confidential, accessible, quiet and suitable for one hybrid participant?

Sensors and telemetry

Room sound statistics, temperature, CO_2 and anonymous occupancy; official lift/door status; aggregate conferencing, network and display status. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Room capacity, privacy features, current bookability and access, accessible route, furniture, hybrid equipment, bookings and permitted use. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: apply privacy, accessibility, capacity and equipment as hard constraints; forecast room conditions and nearby footfall; rank candidates and state which privacy features are documented. Show primary option, backup, trade-offs and evidence time.

Answer boundary

Acoustic measurements do not certify confidentiality. Do not record or analyse mock-viva content. If privacy or hybrid capability is not confirmed by an authoritative source, the room is not a valid recommendation. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Supports a realistic viva rehearsal without compromising confidentiality or accessibility.

PHD-044

CORE | L3 - remote-conference plan | R2 - integration-dependent

PhD student question: I am presenting at an online conference from Abacws. Which room and independent backup plan give me the strongest chance of a reliable, quiet two-hour session?

Sensors and telemetry

Room sound, temperature and CO_2; anonymous occupancy; aggregate network, power and conferencing endpoint status; official planned outage data. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Student access, room bookability, call permission, current equipment, opening hours, network incidents, event time and platform requirements stated by the student. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: current official record or target-room observation with valid spatial coverage. Method: filter for authorised enclosed spaces, assess service health and incident history, forecast room conditions, and provide a primary room, backup room and switch time. Return source and observed-at time; do not infer beyond status.

Answer boundary

A recommendation cannot guarantee an uninterrupted conference session. Use aggregate network/service status only, never inspect traffic or meeting content, and make the backup independent of the same room, circuit or network path where possible. If booking, access or backup service is unverified, identify the unresolved risk explicitly.

Why this is useful: Reduces risk for a time-critical international presentation and makes the fallback explicit.

5. Presentations, conference preparation and viva rehearsal

Questions PHD-045-PHD-046

PHD-045

SITUATIONAL | L2 - poster-production workflow | R2 - integration-dependent

PhD student question: Where can I print, check and assemble a conference poster today without occupying a teaching room or disrupting quiet study?

Sensors and telemetry

Optional anonymous occupancy and broad room condition in assembly areas; authorised print-service, queue and fault status. Sensors cannot determine print entitlement or media stock unless the service exposes it. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; access/lift/door/closure uses authoritative timestamped events.

Authoritative supporting data

Current print facilities, accepted sizes/materials, student entitlement, opening hours, live service status, room purpose, permitted assembly activity, bookings and access. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: authoritative lookup plus calculation or bounded comparison. Minimum evidence: verified permission, access, service, accessibility and timing for every candidate. Method: identify an authorised print route, check service status, select a permitted nearby assembly space and calculate a practical sequence with fallback service instructions. Show trade-offs, an independent backup and uncertainty.

Answer boundary

Do not inspect the poster file or guarantee print quality, stock or queue duration. A clear table is not automatically available for poster assembly. If service or room status is unknown, require direct confirmation. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Helps the doctoral researcher complete a physical conference task without misusing unsuitable spaces.

PHD-046

CORE | L3 - deadline-upload resilience | R2 - integration-dependent

PhD student question: My conference submission is due tonight. Where should I work for the final two hours if I need reliable power, network access and a credible backup upload route?

Sensors and telemetry

Room sound, temperature, CO_2 and anonymous occupancy; aggregate network and power-fault status; UPS-backed service information only from authorised systems. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current student access/opening hours, workspace permission, network and power incident status, institutional submission-system status if exposed, official deadline/time zone supplied by the student and authorised backup submission guidance. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: filter eligible spaces, assess current and historical service incidents, forecast occupancy, rank a primary and physically separate fallback, and set an early switch deadline before submission closes. Show primary option, backup, trade-offs and evidence time.

Answer boundary

The service may recommend a resilient location and backup upload route but cannot guarantee submission acceptance, network continuity or deadline interpretation. It must not inspect the manuscript or credentials. If the authoritative deadline, access or backup path is unclear, ask the student to confirm and avoid a high-confidence recommendation.

Why this is useful: Helps protect a high-consequence submission window against avoidable building-service disruption.

5. Presentations, conference preparation and viva rehearsal

Questions PHD-047-PHD-048

PHD-047

SITUATIONAL | L4 - presentation-condition diagnosis | R3 - research-grade or restricted

PhD student question: My previous presentation room became hot and noisy halfway through. What changed during the session, and how likely is the same pattern next time?

Sensors and telemetry

Distributed room temperature, RH, direct NDIR CO2 and numerical sound statistics; anonymous occupancy and doorway counts; equipment/lighting load; verified HVAC airflow, damper/fan state and outdoor weather. Multiple positions are required if the room has meaningful spatial gradients. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Session time and room supplied by the student, booking, expected attendance if authorised, service incidents and maintenance history. No recording or analysis of presentation content. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: time-aligned target, context and system streams covering baseline, event and recovery. Method: reconstruct the prior session by zone; align occupancy, sound, equipment load, HVAC and outdoor conditions; identify supported contributors and spatial gradients; compare with the next event plan and forecast recurrence with confidence. Test confounders and alternatives; return supported, conflicting or not assessable.

Answer boundary

The reconstruction can identify timing and supported contributors but not prove a single cause. Sound levels do not identify speakers or content. If room sensing is spatially sparse or future attendance/HVAC operation differs, the recurrence forecast must be widened or withheld. If a critical source is missing, stale, conflicting or unauthorised, restrict disclosure, provide only verified facts, or return not assessable.

Why this is useful: Turns a poor presentation experience into evidence for planning the next event without monitoring speech or inventing a cause.

PHD-048

CORE | L4 - viva-room comparison | R2 - integration-dependent

PhD student question: Which candidate room is the stronger choice for my viva with an external examiner when step-free access, confidentiality, hybrid backup and environmental stability all matter?

Sensors and telemetry

Room temperature, CO2, sound and anonymous occupancy; official lift/door status; authorised conferencing, network, display and power status. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current room capacity, privacy and furniture, bookability, access, accessible route, viva requirements, examiner needs voluntarily supplied and institutional room-selection rules. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: overlapping, equivalently placed streams above a declared coverage threshold. Method: enforce policy, accessibility, permission, privacy and equipment as hard constraints; compare historical environmental stability and service reliability; rank candidates with explicit trade-offs and evidence completeness. Report effect size, coverage, uncertainty and sensitivity.

Answer boundary

The building cannot determine academic suitability or replace formal viva arrangements. Acoustic data does not certify confidentiality, and accessibility cannot be inferred from a plan alone. If a hard requirement lacks verification, recommend neither room. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Supports an inclusive and resilient room choice for the most consequential doctoral examination.

5. Presentations, conference preparation and viva rehearsal

Questions PHD-049-PHD-050

PHD-049

SITUATIONAL | L2 - materials route plan | R2 - integration-dependent

PhD student question: I need to carry a poster tube, laptop and demonstration equipment to the presentation room. What is the simplest verified step-free route with the fewest likely pinch points?

Sensors and telemetry

Official lift/door state, corridor footfall and route occupancy; no personal tracking. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; equipment state comes from authorised service logs, not room conditions or power alone.

Authoritative supporting data

Confirmed destination, current accessible route graph, door/lift dimensions, access permissions, temporary closures and any handling restrictions supplied by the student. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: authoritative lookup plus calculation or bounded comparison. Minimum evidence: verified permission, access, service, accessibility and timing for every candidate. Method: remove invalid or restricted paths, estimate current congestion and travel time, identify pinch points and provide a primary and fallback route. Show trade-offs, an independent backup and uncertainty.

Answer boundary

This is not a manual-handling or safety approval. The system cannot guarantee that lifts remain available or corridors remain clear. Official signage, security and event instructions take precedence. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Reduces avoidable delay and handling difficulty immediately before a presentation.

PHD-050

SITUATIONAL | L4 - rehearsal-day optimisation | R2 - integration-dependent

PhD student question: Can you plan a half-day mock-viva rehearsal with a private room, two focused preparation blocks, a quiet break and enough time to reset the room?

Sensors and telemetry

Room sound, temperature, CO_2, illuminance and anonymous occupancy; corridor footfall; aggregate AV/network status and official lift state if required. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Bookings, current room purposes, privacy, permitted use, opening hours, access, equipment and the student's preferred sequence and accessibility needs. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: verified permission, access, service, accessibility and timing for every candidate. Method: allocate preparation, mock viva, break and reset periods; enforce privacy and booking constraints; forecast room conditions and movement; include buffers and fallback spaces. Show trade-offs, an independent backup and uncertainty.

Answer boundary

Do not infer mental state, evaluate performance or record the mock viva. Break suggestions are logistical, not medical. If privacy, booking or room reset permission is unverified, the itinerary must exclude that room. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Makes an intensive rehearsal day manageable while respecting confidentiality and room-use requirements.

6. Late, weekend, commuting and hybrid work

Questions PHD-051-PHD-052

PHD-051

CORE | L1 - weekend access and status | R2 - integration-dependent

PhD student question: I want to work in Abacws on Saturday morning. Which student areas are officially open, accessible to me and currently usable?

Sensors and telemetry

Optional room environmental status and anonymous occupancy; official door, lift, network and service status. Access must not be inferred from occupancy or an open door. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; IT uses authorised timestamped aggregate heartbeat/status only; no traffic, research content or identity.

Authoritative supporting data

Current weekend opening hours, student access permissions, room purpose, closures, bookings, maintenance, security procedure and any restricted-area rules. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: authoritative lookup and/or current observation. Minimum evidence: current official record or target-room observation with valid spatial coverage. Method: list only officially open and authorised areas, attach service and environmental status where verified, and provide official entry and exit guidance. Return source and observed-at time; do not infer beyond status.

Answer boundary

The service cannot grant access, reveal security arrangements or treat presence as permission. Conditions may change after the query; official notices, access controls and staff instructions take precedence. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Prevents a wasted weekend journey and directs the PhD student only to verified usable spaces.

PHD-052

CORE BOUNDARY CASE | L3 - evening lone-work plan | R3 - research-grade or restricted

PhD student question: I expect to work until 9 p.m. What official lone-working steps apply, which workspaces are permitted, and what verified route should I use when leaving?

Sensors and telemetry

Official door/lift state, broad anonymous area occupancy and environmental status; no identity tracking or continuous monitoring of the student. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; IT uses authorised timestamped aggregate heartbeat/status only; no traffic, research content or identity.

Authoritative supporting data

The student's current access entitlement, applicable lone-working policy, building opening/closing times, authorised workspaces, official check-in method, staffed/security support, closures and exit-route status. Sensor presence does not create permission. The official safety source prevails; retain source, issue time, validity window and permission.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: current lone-working rule, access window, staffed/support arrangements, check-in channel, closures and exit routes. Method: filter permitted workspaces and provide required actions, departure trigger, verified route and official escalation contact. Do not use occupancy to infer safety.

Answer boundary

Authorised access does not remove lone-working obligations or guarantee that staff are present. Do not disclose who else is in the building or imply that aggregate occupancy provides personal protection. Official lone-working, check-in, closure and emergency instructions take precedence; if any is unavailable, do not endorse the plan.

Why this is useful: Supports safe, policy-compliant evening work without treating access control or anonymous occupancy as a substitute for lone-working arrangements.

6. Late, weekend, commuting and hybrid work

Questions PHD-053-PHD-054

PHD-053

CORE | L3 - commute-arrival forecast | R2 - integration-dependent

PhD student question: I am travelling in for one long writing session tomorrow. What arrival time gives me the best chance of a suitable workspace without making an unnecessarily early journey?

Sensors and telemetry

Historical and forecast anonymous occupancy, entrance counts, room sound and environmental conditions; corridor footfall. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Opening hours, student spaces, bookings, timetable, events, planned closures, weather and public transport feed; travel preferences voluntarily supplied by the student. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified future status plus comparable quality-controlled history. Method: model workspace demand by arrival time, combine with authoritative travel windows, identify the latest practical arrival with a stated probability and provide an earlier fallback. Report horizon, coverage, prediction interval and recheck trigger.

Answer boundary

Do not infer home location, disability, finances or attendance. Transport and occupancy forecasts are uncertain, and occupancy does not guarantee a desk. If travel or room data are stale, report only verified opening information. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Helps a commuting PhD student balance travel burden against the risk of arriving after suitable spaces fill.

PHD-054

CORE | L4 - hybrid-day itinerary | R2 - integration-dependent

PhD student question: Tomorrow I have an online meeting from home, an afternoon experiment in Abacws and an evening writing block. Can you plan the campus part with realistic buffers?

Sensors and telemetry

Forecast room occupancy, sound and environment; corridor footfall; aggregate network, equipment and lift status for the campus intervals. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Experiment booking and constraints, student access, opening hours, room purpose, permitted activity, equipment status, route graph, planned closures and only the schedule the student provides. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: verified permission, access, service, accessibility and timing for every candidate. Method: anchor the authorised experiment, calculate travel and setup buffers, identify a suitable writing space before or after, forecast conditions and provide fallback routes and rooms. Show trade-offs, an independent backup and uncertainty.

Answer boundary

Do not monitor the home meeting, infer attendance or access unrelated calendar data. The system cannot authorise the experiment or guarantee equipment. If the experiment status is unverified, the campus plan must remain conditional. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Coordinates campus-dependent doctoral work around a hybrid schedule without extending building monitoring into the home.

6. Late, weekend, commuting and hybrid work

Questions PHD-055-PHD-056

PHD-055 SITUATIONAL L3 - weather-disruption routing R2 - integration-dependent

PhD student question: Heavy rain and wind are forecast for my visit. Which authorised entrance and verified step-free route would minimise outdoor exposure and avoid known closures?

Sensors and telemetry

Official door/lift state, corridor footfall and local weather station as context; no inference of route accessibility from environmental sensors alone. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; access/lift/door/closure uses authoritative timestamped events.

Authoritative supporting data

Verified current entrances, access permissions, accessible route graph, covered connections if any, temporary closures, weather warnings and destination. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: filter to authorised step-free routes, compare outdoor exposure and congestion, calculate travel time and provide a fallback if a lift or entrance becomes unavailable. Show primary option, backup, trade-offs and evidence time.

Answer boundary

The service cannot guarantee surface safety, weather conditions or entrance availability. Official severe-weather and campus instructions take precedence. If route accessibility is unverified, do not label it step-free. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Makes an irregular campus visit more manageable during poor weather, especially for students carrying equipment or with access needs.

PHD-056 CORE L3 - disruption impact assessment R2 - integration-dependent

PhD student question: Will planned maintenance, service outages or room closures next week affect my supervision, experiment or usual writing space, and what alternatives should I arrange?

Sensors and telemetry

Optional current environmental and occupancy data for alternatives; official service and closure status is primary. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; equipment state comes from authorised service logs, not room conditions or power alone.

Authoritative supporting data

Planned maintenance, closures, booking changes, access restrictions, supervision and experiment details voluntarily supplied, room inventory, equipment dependencies and opening hours. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: map each planned disruption to the student's stated activities, identify affected dependencies, filter authorised alternatives and provide a time-ordered contingency plan. Show primary option, backup, trade-offs and evidence time.

Answer boundary

Do not infer the student's full schedule or expose other bookings. Planned work can change, and alternative rooms may not reproduce specialist capability. If no validated substitute exists, say so rather than offering a generic room. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Lets the PhD student rearrange scarce supervision or experimental time before a disruption causes failure.

6. Late, weekend, commuting and hybrid work

Questions PHD-057-PHD-058

PHD-057 CORE L1 - late-room condition check R1 - practical with core feeds

PhD student question: It is late and I am already in an authorised workspace. What do current temperature, CO2, lighting and broad anonymous occupancy readings show?

Sensors and telemetry

Correctly mapped in-room temperature, direct NDIR CO2, illuminance and anonymous zone-level occupancy with timestamp, health and gap flags. Corridor readings may be reported separately as context but cannot stand in for the room. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; BMS uses native/event points with a 1-min historian and verified asset-zone command/feedback.

Authoritative supporting data

Authenticated room identity, current student access and official opening status. The system must not infer identity from occupancy sensing. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: authoritative lookup and/or current observation. Minimum evidence: current official record or target-room observation with valid spatial coverage. Method: retrieve the latest quality-checked values, short trend and source freshness; explain that CO2 is a ventilation indicator and occupancy is an anonymous estimate. Return source and observed-at time; do not infer beyond status.

Answer boundary

This is environmental information, not a safety or lone-working clearance. Do not identify anyone present or claim that lighting/CO2 proves the area is safe. If any sensor is stale or outside the room, label that variable unknown. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Provides immediate evidence about the working environment while keeping official after-hours procedures separate.

PHD-058 CORE BOUNDARY CASE L4 - after-hours answerability R3 - research-grade or restricted

PhD student question: Using only privacy-preserving aggregate sensing, can the system tell whether this authorised zone may be sparsely occupied, and what do the official lone-working rules require?

Sensors and telemetry

Privacy-preserving zone-level occupancy sensing only: anonymous mmWave/PIR presence or doorway counts aggregated over a declared interval, with coverage, count-error, timestamp, uptime and health flags. No cameras, device identifiers, speech, individual trajectories or exact person-location disclosure. Environmental readings may provide context but do not establish whether the student is alone.

Authoritative supporting data

Authoritative after-hours access and lone-working policy, zone definition, opening/closure status and the student's authenticated access entitlement. Occupancy estimates remain separate from access logs; exact identities and other users' movements are not disclosed. Preserve source validity and permissions.

Analysis required

Operation: authorised policy lookup plus privacy-preserving estimate. Minimum evidence: current lone-working rules and a commissioned aggregate occupancy feed with known spatial coverage and error. Method: report only a coarse occupancy class such as no activity detected, sparse activity possible or activity detected; state timestamp and uncertainty; then present the official required actions. Never infer that the student is safe because activity is detected.

Answer boundary

The system must not state that the student is the only person present, identify anyone, or use access logs to reveal other occupants. A no-activity estimate may be a false negative and is not a safety clearance. If the policy or aggregate sensing is unavailable, provide official contact routes and mark occupancy status unknown.

Why this is useful: Clarifies after-hours obligations without turning aggregate occupancy sensing into individual surveillance or a safety guarantee.

6. Late, weekend, commuting and hybrid work

Questions PHD-059-PHD-060

PHD-059

SITUATIONAL | L2 - short-visit plan | R2 - integration-dependent

PhD student question: I only need to collect approved equipment and leave. What is the shortest authorised route, and is the destination officially accessible now?

Sensors and telemetry

Official door/lift state and corridor footfall; optional destination environmental status if equipment-sensitive. No personal tracking. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; IT uses authorised timestamped aggregate heartbeat/status only; no traffic, research content or identity.

Authoritative supporting data

Equipment collection authorisation, destination, current access, opening hours, route graph, planned closures and any handover procedure. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: authoritative lookup plus calculation or bounded comparison. Minimum evidence: current official record or target-room observation with valid spatial coverage. Method: verify that collection is authorised and the destination is open, compute the shortest valid route, estimate travel time and provide a fallback. Return source and observed-at time; do not infer beyond status.

Answer boundary

The system cannot release equipment, bypass sign-out or disclose restricted storage. An open route does not prove the item is ready. If collection status or access is unverified, direct the student to the responsible service. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Avoids a wasted journey and unnecessary time in the building for a narrowly defined task.

PHD-060

SITUATIONAL | L3 - weekend compute-service check | R2 - integration-dependent

PhD student question: I want to start an authorised long-running compute job over the weekend. Are the managed service and required infrastructure available, and what official failure notifications can I rely on?

Sensors and telemetry

For managed on-site computing only: authoritative scheduler/service heartbeat, queue and storage-health summaries, UPS/PDU and cooling status for the verified hosting facility, plus room/rack temperature and leak alarms where officially exposed. Environmental sensors in the student workspace are irrelevant unless the job runs on local authorised equipment. Preserve timestamps, service scope and health flags; inspect no code, data or network content.

Authoritative supporting data

Approved managed-compute service catalogue, user entitlement, maintenance window, service owner, queue/power/cooling status, permitted unattended operation, storage/checkpoint rules and official alert channel. Room access does not authorise compute use. The official safety source prevails; retain source, issue time, validity window and permission.

Analysis required

Operation: authoritative service lookup plus bounded readiness estimate. Minimum evidence: confirmed entitlement, maintenance window, service heartbeat and failure-notification route; infrastructure telemetry only for the verified hosting facility. Method: report current service state, scheduled outages, queue or capacity indicators and notification coverage; distinguish official availability from any estimate of completion risk. Do not predict scientific runtime without user-supplied job characteristics.

Answer boundary

The assistant cannot approve the job, inspect its content, guarantee completion or override scheduler, storage or security policy. Room conditions are relevant only to local authorised hardware. If service scope, entitlement, maintenance or notification data is unavailable, direct the student to the managed-compute service and mark readiness unknown.

Why this is useful: Helps plan weekend computing around verified service availability and notification routes without exposing research content.

7. Accessibility, well-being, lone-working and safety

Questions PHD-061-PHD-062

PHD-061 CORE L3 - accessible low-crowding route R2 - integration-dependent

PhD student question: I need a step-free route to supervision that avoids the busiest and noisiest areas around class changeover. Which verified route should I take?

Sensors and telemetry

Official lift/door status, corridor and stair footfall, numerical acoustic statistics and temporary-route status. No identity tracking. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current accessible route graph, supervision destination/time supplied by the student, entrance permissions, closures and any route preferences voluntarily stated. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: current official record or target-room observation with valid spatial coverage. Method: remove non-step-free or unverified segments, forecast changeover congestion and sound, rank valid routes by accessibility, crowding and travel time, and provide a fallback. Return source and observed-at time; do not infer beyond status.

Answer boundary

Do not infer disability or disclose other users. Sound and crowding estimates do not make an otherwise inaccessible route valid. Lift and closure status can change, so the route requires a refresh before departure. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Supports reliable access to supervision while respecting sensory and mobility preferences.

PHD-062 CORE L3 - accessible-workspace recommendation R2 - integration-dependent

PhD student question: Which current student workspace has verified step-free access, suitable adjustable furniture, power and a comparatively calm environment for a long writing session?

Sensors and telemetry

Room sound, temperature, CO_2 and illuminance; anonymous occupancy; official lift/door status and electrical-fault status. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current accessible-space inventory, verified furniture and desk features, access, bookability, room purpose, power provision, opening hours and closures. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: enforce accessibility, furniture, access and power as hard constraints; compare present and forecast room conditions; rank primary and backup spaces and show which features are verified. Show primary option, backup, trade-offs and evidence time.

Answer boundary

Do not infer a medical condition or assume that nominally adjustable furniture suits every user. Environmental calm is a forecast, not a guarantee. If furniture or route information is stale, do not present the space as accessible. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Helps a doctoral student find a workspace that supports both physical access and sustained research work.

7. Accessibility, well-being, lone-working and safety

Questions PHD-063-PHD-064

PHD-063

CORE | L3 - low-stimulation space recommendation | R2 - integration-dependent

PhD student question: I need a low-stimulation place for two hours, with low foot traffic, stable lighting and low background sound. What is the best verified option?

Sensors and telemetry

Corridor footfall, anonymous room occupancy, locally calculated sound statistics and task-area illuminance; temperature and CO_2 for general condition. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current student access, room purpose, bookings, opening hours, lighting controls and route accessibility. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: filter permitted spaces, compare sound, footfall and lighting stability, forecast the requested period and rank options with a fallback. Show primary option, backup, trade-offs and evidence time.

Answer boundary

The service must not diagnose sensory conditions or promise a disturbance-free environment. A single lux sensor may not capture glare at every seat. Ask for clarification if the user has a specific access requirement that the data cannot represent. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Supports focused or recovery time while keeping the recommendation environmental rather than medical.

PHD-064

SITUATIONAL | L3 - glare-aware recommendation | R2 - integration-dependent

PhD student question: Glare is making it difficult to read my screen. Is another nearby permitted workspace likely to have more stable light for the next two hours?

Sensors and telemetry

Task-plane illuminance, daylight/solar reference, optional window or blind state where authorised, room occupancy and temperature. Sensor geometry and calibration must be documented. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; access/lift/door/closure uses authoritative timestamped events.

Authoritative supporting data

Current room access and use, verified nearby workspaces, bookings, route and any controllable-lighting or blind information. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: assess whether measured light is changing rapidly, compare candidate spaces and solar exposure, and rank a nearby alternative with uncertainty. Show primary option, backup, trade-offs and evidence time.

Answer boundary

Illuminance alone does not measure screen glare or visual comfort for an individual. Do not make a medical claim. If sensor placement does not represent the workstation or blind state is unknown, describe the evidence as indirect. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Gives the PhD student a practical option when lighting conditions disrupt screen-based doctoral work.

7. Accessibility, well-being, lone-working and safety

Questions PHD-065-PHD-066

PHD-065

SITUATIONAL | L3 - long-session break planning | R2 - integration-dependent

PhD student question: I have a long writing day planned. Can you suggest optional break times and nearby break locations without monitoring or judging my productivity?

Sensors and telemetry

Room occupancy, sound, temperature, CO_2 and nearby corridor footfall; optional daylight and outdoor weather for break-area context. No personal activity monitoring. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Current student-accessible break areas, opening hours, route graph and the work interval the student chooses to provide. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: divide the stated work period into optional break opportunities, identify nearby permitted spaces with suitable current conditions and route time, and offer choices rather than behavioural directives. Show primary option, backup, trade-offs and evidence time.

Answer boundary

The system must not infer fatigue, concentration or adherence and must not provide medical advice. Break timing is a user-controlled planning aid. If no verified break area exists, provide only route and opening information. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Supports sustainable long-duration working while explicitly rejecting productivity surveillance.

PHD-066

CORE BOUNDARY CASE | L4 - environment-and-symptom evidence review | R3 - research-grade or restricted

PhD student question: I often develop a headache in this room. Can you show whether any measured environmental pattern coincides with the times I choose to report, without diagnosing me?

Sensors and telemetry

Correctly located temperature, RH, direct NDIR CO_2, validated PM2.5 where installed, illuminance/glare context and numerical sound statistics; sensor quality metadata; only symptom times deliberately entered for this query. No passive health inference or individual tracking. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; BMS uses native/event points with a 1-min historian and verified asset-zone command/feedback.

Authoritative supporting data

Current room identity and use, maintenance or cleaning events, outdoor air and HVAC status. Personal symptom entries require explicit consent, minimisation and deletion controls. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: time-aligned target, context and system streams covering baseline, event and recovery. Method: use only voluntarily supplied symptom times; compare them with matched control periods; calculate differences, coverage and sensor uncertainty; present associations and alternative explanations without causal or medical inference. Test confounders and alternatives; return supported, conflicting or not assessable.

Answer boundary

This is a user-initiated environmental correlation, not a medical assessment. Do not infer health, diagnosis, disability or productivity, and retain only the minimum voluntarily supplied times under an appropriate consent and retention rule. If timing, room-level coverage or confounders are inadequate, show measurements only and advise the student to follow appropriate health or university support routes.

Why this is useful: Lets the student examine a self-reported pattern responsibly while keeping medical interpretation outside the building system.

7. Accessibility, well-being, lone-working and safety

Questions PHD-067-PHD-068

PHD-067

CORE | L1 - lone-working policy lookup | R2 - integration-dependent

PhD student question: What official lone-working rules apply to my planned evening experiment, and what approval, check-in or supervision do I need?

Sensors and telemetry

None required to determine policy; official building status and authorised equipment alarms may be displayed as secondary context. Acquisition and privacy: equipment state comes from authorised service logs, not room conditions or power alone.

Authoritative supporting data

Approved risk assessment and method, lone-working policy, current room/equipment authorisation, induction, supervision requirement, official check-in/escalation process, certified alarms and responsible-person decision. The official safety source prevails; retain source, issue time, validity window and permission.

Analysis required

Operation: authoritative lookup and/or current observation. Minimum evidence: current official policy, alarm or approval; sensor streams are supplementary. Method: identify the applicable rule from the authenticated activity and location, list required approvals/check-ins and direct the student to the competent authority for unresolved conditions. Report source and issue time; never infer clearance or emergency action.

Answer boundary

The conversational system cannot approve lone working, waive supervision or interpret a risk assessment beyond the authoritative text. Sensor presence does not make an activity safe. If activity or location is ambiguous, require clarification. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Gives the PhD student a clear route to the official requirements before committing to after-hours research work.

PHD-068

CORE BOUNDARY CASE | L1 - authoritative closure or emergency response | R2 - integration-dependent

PhD student question: An alarm or urgent official building message has appeared. What is the current authorised instruction, and which authoritative system issued it?

Sensors and telemetry

No experimental sensor is authoritative for the instruction. Use certified alarm/status interfaces only where officially integrated; environmental or occupancy telemetry may be shown to authorised responders as supplementary context, never as a substitute for the alarm or official message.

Authoritative supporting data

Authoritative sources only: certified alarm state and official institutional emergency/closure message, with timestamp, source, authorised route or contact information and current on-site instructions. Experimental sensors are secondary context and must not generate the instruction. The official safety source prevails; retain source, issue time, validity window and permission.

Analysis required

Operation: authoritative emergency lookup. Minimum evidence: a current, authenticated official alert with issuing system, affected scope, timestamp and status. Method: reproduce or faithfully summarise the active instruction, identify the source and time, suppress conflicting non-authoritative suggestions, and direct the user to follow signage, wardens, security or emergency services. No independent diagnosis is performed.

Answer boundary

Do not reinterpret, delay or contradict an official alarm or emergency instruction. The conversational service must not calculate an evacuation route that conflicts with live authoritative directions, disclose sensitive security information or imply that silence means safety. If the official feed is unavailable, tell the user to follow on-site alarms, signage and emergency procedures.

Why this is useful: Provides one clear, authoritative source of emergency instruction and prevents sensor-derived speculation from competing with it.

7. Accessibility, well-being, lone-working and safety

Questions PHD-069-PHD-070

PHD-069

CORE BOUNDARY CASE | L4 - experiment authorisation boundary | R3 - research-grade or restricted

PhD student question: Do current room readings provide any safety clearance for running this experiment alone tonight, or must the decision come from the approved risk assessment and responsible person?

Sensors and telemetry

Only hazard-specific, risk-assessed and certified monitoring can contribute to a safety decision. Generic temperature, CO2, PM or occupancy sensors may provide context but cannot authorise an experiment, replace supervision, verify containment or override certified alarms and interlocks. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; equipment state comes from authorised service logs, not room conditions or power alone.

Authoritative supporting data

Approved risk assessment, method statement, lone-working and supervision rules, induction, room/equipment authorisation, interlock/certified-system status and explicit approval from the responsible person where required. The official safety source prevails; retain source, issue time, validity window and permission.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: approved risk assessment, permitted lone-working conditions, required supervision/check-in, equipment status and certified controls. Method: use room telemetry only as contextual observation and return permission confirmed by the responsible process, permission absent, or not assessable - never sensor-derived safety clearance.

Answer boundary

The answer cannot be a safety clearance. Environmental readings, occupancy estimates and equipment status are supplementary only; approval must come from the current risk assessment, lone-working procedure and responsible competent person. If any required approval or certified hazard control is absent, the only responsible answer is that the experiment must not be treated as authorised.

Why this is useful: Prevents environmental telemetry from being mistaken for permission or safety clearance for lone experimental work.

PHD-070

SITUATIONAL | L3 - supported after-hours workspace recommendation | R3 - research-grade or restricted

PhD student question: If I have authorised evening access, which permitted workspace keeps me near an officially staffed or supported area without revealing who else is in the building?

Sensors and telemetry

Privacy-preserving zone-level occupancy bands, official staffed-area status, door/lift state and room environmental condition. No identities or precise locations of other occupants. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; access/lift/door/closure uses authoritative timestamped events.

Authoritative supporting data

Current access, staffed or security-supported zones, lone-working policy, opening hours, room use, closures and route graph. The official safety source prevails; retain source, issue time, validity window and permission.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: current official policy, alarm or approval; sensor streams are supplementary. Method: filter authorised spaces, prioritise proximity to officially supported areas and valid exits, use only broad occupancy as context, and provide the required check-in procedure. Report source and issue time; never infer clearance or emergency action.

Answer boundary

Do not imply that nearby anonymous occupancy guarantees assistance or disclose where individuals are. Official lone-working rules remain binding. If staffed-area status is unavailable, do not make a support-based recommendation. If a critical source is missing, stale, conflicting or unauthorised, restrict disclosure, provide only verified facts, or return not assessable.

Why this is useful: Supports a more informed evening workspace choice without compromising other occupants' privacy.

8. Sustainability, evidence quality and responsible research use

Questions PHD-071-PHD-072

PHD-071

SITUATIONAL | L3 - low-impact workspace recommendation | R2 - integration-dependent

PhD student question: For a four-hour writing session, which available workspace is the lower-energy option while still meeting reasonable ventilation, lighting, temperature and accessibility needs?

Sensors and telemetry

Room/zone lighting and HVAC operating state, anonymous occupancy, direct NDIR CO2, temperature, illuminance and suitably allocated electrical or thermal energy where metering permits. Whole-zone energy must not be presented as exact marginal energy for one student. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; BMS uses native/event points with a 1-min historian and verified asset-zone command/feedback.

Authoritative supporting data

Current access, room use, bookings, opening hours, metering and HVAC-zone relationships, energy data quality and any institutional operating constraints. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: estimate or forecast followed by a constrained recommendation. Minimum evidence: verified eligibility and status for every candidate, with target-room evidence where used. Method: filter by current access, activity, accessibility and environmental minimums; estimate only the energy component supported by meter boundaries; compare options for the requested period; rank alternatives and expose energy/comfort trade-offs and uncertainty. Show primary option, backup, trade-offs and evidence time.

Answer boundary

A low-energy ranking is only valid within verified meter boundaries and cannot outweigh accessibility, ventilation, lighting or permitted use. Do not claim exact marginal energy for one person from shared-zone data. Missing room-level coverage or access data requires a limited answer. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Supports a lower-impact workspace choice without sacrificing accessibility, indoor environmental quality or authorised use.

PHD-072

ADVANCED | L4 - energy-normalised space comparison | R3 - research-grade or restricted

PhD student question: Among comparable PhD workspaces, which use less building energy per estimated occupied hour while maintaining similar environmental conditions?

Sensors and telemetry

Comparable room/zone energy submetering, anonymous occupancy estimates with known error, room size/use and HVAC service relationships, temperature, direct NDIR CO2, illuminance, weather and schedule context. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; BMS uses native/event points with a 1-min historian and verified asset-zone command/feedback.

Authoritative supporting data

Verified room type and capacity, metering boundaries, HVAC service relationships, operating schedules, room-use changes and maintenance history. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: verified meter and occupancy boundaries, overlapping comparable periods and declared coverage. Method: calculate energy per estimated occupied hour with occupancy-error range; normalise for weather, schedule and shared HVAC only where defensible. Report sensitivity and unallocatable cases.

Answer boundary

Energy per estimated occupied hour can be highly uncertain when counts are noisy or zones share services. Do not treat the result as an individual footprint or a causal efficiency rating. Exclude non-comparable spaces and report sensitivity to occupancy error. If a critical source is missing, stale, conflicting or unauthorised, restrict disclosure, provide only verified facts, or return not assessable.

Why this is useful: Enables research-grade comparison of space-energy performance while exposing the uncertainty introduced by shared zones and estimated occupancy.

8. Sustainability, evidence quality and responsible research use

Questions PHD-073-PHD-074

PHD-073 SITUATIONAL L4 - empty-space conditioning analysis R3 - research-grade or restricted

PhD student question: Are any student workspaces repeatedly heated, cooled, ventilated or lit when aggregate evidence suggests they are unoccupied, and how reliable is that conclusion?

Sensors and telemetry

Anonymous occupancy estimates, lighting state and illuminance, HVAC airflow/valve/damper/fan state, zone energy where available, timetable/bookings and known after-hours control requirements. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; BMS uses native/event points with a 1-min historian and verified asset-zone command/feedback.

Authoritative supporting data

Room purpose, bookings, operating schedules, required ventilation or equipment loads, cleaning/maintenance activity and HVAC topology. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: time-aligned occupancy/booking evidence and HVAC/lighting/energy state with verified zone boundaries. Method: detect sustained vacancy-operation mismatches; exclude documented start-up, setback recovery, cleaning, frost protection and shared-zone service. Quantify duration and uncertainty, not automatic waste.

Answer boundary

Apparent vacancy may reflect sensor undercounting, cleaning, scheduled preconditioning, frost protection or service to another room. Do not automatically label operation waste. If shared-zone energy cannot be allocated defensibly, report duration of apparent mismatch without a cost claim. If a critical source is missing, stale, conflicting or unauthorised, restrict disclosure, provide only verified facts, or return not assessable.

Why this is useful: Identifies potentially avoidable operation for investigation without wrongly labelling legitimate building control as waste.

PHD-074 CORE L4 - responsible data-access assessment R3 - research-grade or restricted

PhD student question: I want to use Abacws data in my thesis. Which datasets may I request, for what purposes, and what governance, licensing or ethics conditions apply?

Sensors and telemetry

Dataset inventory and metadata rather than additional sensing; provenance, aggregation level, personal-data risk, retention, licence and access-control classification. Acquisition and privacy: access/lift/door/closure uses authoritative timestamped events.

Authoritative supporting data

Approved dataset catalogue, data owner, licence and permitted purpose, lawful basis/consent where applicable, ethics requirement, security classification, retention/export rules, data-management policy and formal access-request route. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: units, processing version, provenance and declared valid-data coverage. Method: match the proposed research purpose to permitted datasets, identify approvals and minimisation requirements, distinguish open, de-identified, restricted and unavailable data, and provide the formal request route. Expose formula or decision rule, exclusions, uncertainty and source time.

Answer boundary

The building can explain discoverability and the request route but cannot grant access, waive ethics, licences or data-protection duties, or infer that de-identification removes all risk. Do not expose restricted telemetry or personal data. If purpose, legal basis, licence, retention, security or approval is unresolved, classify access as pending or not permitted rather than supply the dataset.

Why this is useful: Shows the legitimate route from data discovery to governed research access, reducing ethical and licensing mistakes.

8. Sustainability, evidence quality and responsible research use

Questions PHD-075-PHD-076

PHD-075

CORE | L2 - provenance and calibration explanation | R2 - integration-dependent

PhD student question: For this temperature-and-CO2 result, can you show exactly which sensors, locations, calibration records, exclusions, transformations and time periods were used?

Sensors and telemetry

Sensor identity, observed property, units, precise location/served zone, mounting, make/model, stated accuracy, calibration or commissioning, sampling and aggregation, timestamp quality, health/gap flags, time-series mapping, transformations, exclusions and source-to-result lineage. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: equipment state comes from authorised service logs, not room conditions or power alone.

Authoritative supporting data

Ontology and sensor registry, deployment/maintenance/calibration records, data-processing and code version, inclusion/exclusion log, access permissions and any manual correction record. Publication in a paper does not by itself establish current access or completeness. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: provenance lookup plus reproducibility calculation. Minimum evidence: immutable sensor and ontology identifiers, location/served-zone history, units, sampling and aggregation, calibration/maintenance, data-quality flags, exclusions, transformations, code or query version and the exact interval. Method: trace each reported value back to source observations, quantify coverage and exclusions, and distinguish direct readings from derived metrics. Output: a machine-readable and human-readable provenance trail.

Answer boundary

Do not expose credentials, restricted infrastructure details, booking identities or personal data. Missing calibration limits the claim but does not automatically invalidate every reading. If source-to-result lineage is incomplete, state that the result is not fully reproducible. If a hard constraint or required room-level source is missing, stale or conflicting, provide only confirmed facts or return not assessable.

Why this is useful: Provides the traceability the PhD student needs to reproduce, critique and appropriately qualify a reported result.

PHD-076

CORE | L4 - missing-data sensitivity analysis | R3 - research-grade or restricted

PhD student question: How much does my room-comparison conclusion change if I exclude periods with sensor gaps or use another scientifically justified missing-data treatment?

Sensors and telemetry

Source time series, explicit missingness and quality flags, calibration/relocation history and comparable reference streams. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: BMS uses native/event points with a 1-min historian and verified asset-zone command/feedback.

Authoritative supporting data

Data dictionary, processing pipeline, room/HVAC relationships, study period and analysis choices supplied by the student. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: identical spatial/temporal scope, declared coverage and pre-specified defensible gap treatments. Method: rerun the comparison under complete-case, exclusion or justified imputation rules; report effect size, uncertainty and whether the conclusion changes. Preserve every rule and output.

Answer boundary

The system must not select an imputation method without the study assumptions or hide non-random missingness. Interpolation across long gaps, relocation or sensor replacement may be invalid. Do not claim robustness when only one defensible treatment exists. If a critical source is missing, stale, conflicting or unauthorised, restrict disclosure, provide only verified facts, or return not assessable.

Why this is useful: Shows whether a thesis result is robust to defensible data-cleaning choices rather than dependent on one convenient treatment.

8. Sustainability, evidence quality and responsible research use

Questions PHD-077-PHD-078

PHD-077

CORE | L2 - spatial answerability query | R1 - practical with core feeds

PhD student question: Can the corridor sensor outside this office tell me what conditions were like inside, or is valid room-level evidence required?

Sensors and telemetry

The corridor sensor and its exact location plus any in-room or validated served-zone sensor, door state and HVAC relationships. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: BMS uses native/event points with a 1-min historian and verified asset-zone command/feedback; access/lift/door/closure uses authoritative timestamped events.

Authoritative supporting data

Current spatial model, sensor registry, room boundary, ventilation relationships and deployment records. Use records valid for the requested interval and preserve source owner/version, retrieval time and permission; missing or denied fields remain unknown.

Analysis required

Operation: authoritative lookup plus calculation or bounded comparison. Minimum evidence: units, processing version, provenance and declared valid-data coverage. Method: determine whether the sensor is inside the target environmental boundary or a validated served zone, explain possible separation by doors and airflow, and classify the claim as direct, indirect context or not assessable. Expose formula or decision rule, exclusions, uncertainty and source time.

Answer boundary

A nearby corridor sensor may be reported only as contextual evidence with its distance, separation and uncertainty. It cannot be relabelled as an office measurement or used for room-level compliance, comfort or thesis claims without validated transfer evidence. If no valid in-room or served-zone measurement exists, the room-level answer is not assessable.

Why this is useful: Prevents a convenient nearby measurement from becoming an invalid room-level claim in research or facilities reporting.

PHD-078

CORE | L2 - privacy and data-minimisation query | R2 - integration-dependent

PhD student question: What can the building infer from occupancy and acoustic sensing, and what technical and governance controls prevent individual monitoring?

Sensors and telemetry

Occupancy/acoustic sensor type, field of view or spatial granularity, feature extraction, aggregation, retention, access logging and combination risk. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: sound uses 1-s local features retained as 1-min LAeq/L10/L90 only; no audio or speech; occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds.

Authoritative supporting data

Privacy notice, DPIA or equivalent assessment, retention schedule, role-based access policy, audit process, ethics conditions for secondary research use and data-subject rights. Missing governance documentation means protections cannot be verified. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: authorised privacy and capability lookup. Minimum evidence: documented sensing modalities, on-device processing, retained fields, aggregation, access control, retention and DPIA/governance records. Method: list what each modality can directly observe, what the system deliberately discards, which inferences are prohibited, who can access aggregates and how a user can challenge or report misuse. No inference about current occupants is performed.

Answer boundary

Do not demonstrate privacy by revealing live identities, trajectories, conversations or precise occupancy. Claims about anonymity must reflect the actual implementation and residual re-identification risk, not marketing language. If documentation or controls are incomplete, state the limitation and do not claim that individual monitoring is impossible.

Why this is useful: Makes the sensing system's privacy limits understandable without revealing information about current occupants.

8. Sustainability, evidence quality and responsible research use

Questions PHD-079-PHD-080

PHD-079 ADVANCED L4 - lower-carbon compute scheduling R3 - research-grade or restricted

PhD student question: For a non-urgent authorised compute job, is there a lower-carbon time to run it without increasing failure risk or missing a fixed deadline?

Sensors and telemetry

Authorised compute queue/load and energy telemetry, facility power and cooling status, local renewable generation if present, and external grid carbon-intensity forecast from an authoritative source. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: BMS uses native/event points with a 1-min historian and verified asset-zone command/feedback; meters use verified boundaries with 1-min demand and 15-min energy.

Authoritative supporting data

Compute-service policy, job priority and deadline supplied by the student, maintenance windows, carbon-intensity feed, power/cooling incidents and any institutional carbon methodology. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: diagnosis, optimisation, longitudinal comparison or evidence audit. Minimum evidence: workload flexibility, deadline, checkpoint capability, approved service, queue/maintenance, power/cooling risk and authorised carbon-intensity source. Method: forecast feasible windows and rank them by carbon difference and failure risk. Report uncertainty and do not delay fixed-deadline or safety-critical work.

Answer boundary

This is a scheduling recommendation, not a guarantee of lower marginal emissions or successful completion. Use approved carbon-intensity and service data, preserve deadlines and reliability constraints, and inspect no job content. If the compute platform cannot defer safely or carbon data are stale, recommend the earliest reliable authorised window or mark the carbon comparison not assessable.

Why this is useful: Allows the PhD student to consider the environmental timing of flexible computation without compromising research deadlines or service reliability.

PHD-080 CORE L4 - thesis evidence package R3 - research-grade or restricted

PhD student question: Can you assemble a reproducible evidence package for my thesis claim about ventilation or thermal conditions in a specified Abacws space and period?

Sensors and telemetry

Quality-controlled room sensors appropriate to the claim, verified BMS/HVAC points when ventilation is inferred, outdoor reference and occupancy context where needed, plus complete calibration, location, maintenance, missingness, processing-version and permission metadata. Evidence quality: verified target-room/served-zone mapping, synchronised observed-at times, commissioning/calibration, uptime, gap, drift and source-agreement flags. Acquisition and privacy: occupancy uses event/1-5 s anonymous sensing aggregated to 1-min counts with coverage and error bounds; BMS uses native/event points with a 1-min historian and verified asset-zone command/feedback.

Authoritative supporting data

Current spatial/HVAC model, sensor registry, data dictionary, licence and access approval, room-use and maintenance history, research definition/criterion supplied by the student, and versioned analysis method. The assistant cannot choose the thesis claim. Preserve source owner, version, validity, permission and audit trail; discoverability is not authority to disclose or reuse.

Analysis required

Operation: evidence audit and reproducible package assembly. Minimum evidence: a precisely scoped claim, verified room/served-zone mapping, quality-controlled observations for the stated interval, provenance, processing version, uncertainty and governing interpretation. Method: compile source identifiers, units, timestamps, calibration/maintenance, coverage, exclusions, transformations, sensitivity checks and conflicting evidence; label every statement as observed, calculated or inferred. Output: citable tables plus a machine-readable manifest.

Answer boundary

The package supports transparent analysis; it does not certify legal compliance, causal interpretation or scientific validity beyond the stated evidence. Do not include restricted or personal data without approval. If spatial coverage, provenance, calibration, permissions or a reproducible processing chain is inadequate, identify the gap and mark the corresponding claim not assessable.

Why this is useful: Creates a transparent, reproducible basis for a thesis claim while exposing uncertainty, conflicts and evidence gaps.

4. Catalogue summary and implementation requirements

The catalogue combines practical PhD-student service questions with integration-dependent planning, research-grade evidence assessments and restricted safety cases. Readiness identifies the verified sensing, source authority, quality threshold, permissions and safe-failure controls required for a responsible answer.

Questions	80	Eight PhD-student-centred sections containing ten questions each.
R1 practical	2	Can be supported with proposed core sensing and basic authoritative spatial or service feeds.
R2 integration-dependent	52	Require booking, access, doctoral, IT, AV, lift, BMS, equipment, maintenance or forecasting integration.
R3 research or restricted	26	Require dense sensing, validated diagnosis, controlled research or operational data, or stronger privacy, ethics and safety governance.

Recurring system capabilities

1 Long-horizon and irregular planning

Resolve recurring writing windows, supervision, experiments, milestones, commuting, hybrid and after-hours constraints across long doctoral timescales.

2 Room, route and permission reasoning

Use current room purpose, PhD-student access, bookings, capacities, specialist authorisation, step-free routes, lift status and closures.

3 Environmental, equipment and service evidence

Combine correctly located room sensors with anonymous occupancy, BMS, energy, network, AV, compute, equipment and authoritative safety status.

4 Forecast, comparison and diagnosis

Predict demand and conditions, build matched baselines, rank alternatives, reconstruct events and separate association from supported diagnosis.

5 Research evidence quality and provenance

Check room/served-zone mapping, sampling and aggregation, clocks, commissioning/calibration, uptime, gaps, drift, source agreement, processing lineage, permissions and whether each claim is direct or inferred.

6 Privacy, ethics, safety and abstention

Minimise personal and research data, refuse unauthorised access or safety clearances, ask for missing constraints and return not assessable when evidence is inadequate.

QUESTION RECORD STRUCTURE

Each entry can become a system test containing the natural utterance and paraphrases, required entities, minimum room and service evidence, source-precedence rule, quality threshold, expected operation type, permission rule, answerability class, acceptable clarification, uncertainty statement, safe failure, expected answer characteristics and prohibited overclaims.

5. Evidence, permissions and answerability rules

These rules apply across all 80 records and define when the building may answer, qualify, ask for clarification or return not assessable.

Building and room status

The May 2021 plan is a historical spatial reference. Current room purpose, ownership, capacity, PhD-student access, bookability, specialist facilities, equipment, entrances, lifts, services and closures require an authoritative current source.

Spatial coverage

A room-level answer requires a correctly located sensor inside that room or a validated served zone. Corridor and adjacent-room data may be labelled context but cannot be silently substituted.

Time window and operation type

Every answer resolves the requested interval and labels its operation as observation, authoritative lookup, calculation, estimate, forecast, diagnosis or recommendation. Preserve observed-at and retrieved-at times; stale evidence is not current status.

Calibration and data quality

Use stated sampling and aggregation, synchronised observed-at times, commissioning/calibration, uptime, gap and drift flags, maintenance and source agreement. A failed, stale, relocated or unverified point cannot support a precise comparison, forecast, diagnosis or thesis claim.

Research-content minimisation

Do not inspect drafts, code, datasets, experiments, presentations, conversations or network traffic. Process only the minimum user-supplied schedule, activity and requirements needed for the request.

Occupancy, sound and privacy

Use anonymous aggregate counts and numerical LAeq or percentile statistics. Do not identify people, retain speech, infer conversation content, track movement or estimate individual productivity.

Environmental interpretation

Direct-reading NDIR CO_2 is a ventilation indicator, not complete air quality. Temperature, RH, PM, indicative TVOC, airflow, radiant conditions, outdoor weather and activity may be relevant subject to coverage and quality.

Experiments and specialist spaces

Environmental IoT data cannot authorise research activity or certify safety. Current room function, induction, supervision, risk assessment, equipment, compute entitlement and official procedures come from authoritative systems.

Operational services and permissions

Supervision details, bookings, access, lifts, network, AV, compute, printing, maintenance and closures come from authorised feeds. Enforce the PhD-student role and minimise personal and schedule data.

Emergency and high-consequence answers

Certified alarms and official communications take precedence. Safety, health, lone-working, ethics and compliance questions require competent authority; missing or conflicting critical evidence requires a limited or not-assessable answer.

FINAL ANSWERABILITY RULE

Answer only when the required variables, current room/served-zone coverage, requested interval, source authority, quality threshold, permissions and safety constraints are adequate. Otherwise ask for clarification, omit unsupported criteria, return confirmed facts with explicit uncertainty, restrict disclosure, or mark the analytical part not assessable.